

## FORECAST OF ATLANTIC SEASONAL HURRICANE ACTIVITY AND LANDFALL STRIKE PROBABILITY FOR 2026

We have maintained our 2026 Atlantic basin hurricane season forecast for a well below-normal season. El Niño conditions are likely to continue to intensify over the next few months. We are extremely confident that we will have a strong El Niño for the peak of hurricane season. Sea surface temperatures across the Caribbean and tropical Atlantic are slightly above their long-term averages. We anticipate the powerful El Niño being the dominant factor for the upcoming hurricane season, driving very high levels of tropical Atlantic vertical wind shear. We are forecasting a well below-average probability for major hurricane landfalls along the continental United States coastline and in the Caribbean. As with all hurricane seasons, coastal residents are reminded that it only takes one hurricane making landfall to make it an active season. Thorough preparations should be made every season, regardless of predicted activity.

(as of 5 August 2026)

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With Special Assistance from the TC-RAMS Team<sup>6</sup> and Carl J. Schreck III<sup>7</sup>  
In Memory of William M. Gray<sup>8</sup>

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## ATLANTIC BASIN SEASONAL HURRICANE FORECAST FOR 2026

| Forecast Parameter and 1991-2020<br>Average (in parentheses) | Issue Date<br>9 April<br>2026 | Issue Date<br>10 June<br>2026 | Issue Date<br>8 July<br>2026 | Issue Date<br>5 August<br>2026 | Observed Thru<br>4 August<br>2026 | Remainder of<br>Season<br>Forecast |
|--------------------------------------------------------------|-------------------------------|-------------------------------|------------------------------|--------------------------------|-----------------------------------|------------------------------------|
| Named Storms (NS) (14.4)                                     | 13                            | 11                            | 9                            | 9                              | 2                                 | 7                                  |
| Named Storm Days (NSD) (69.4)                                | 55                            | 45                            | 35                           | 35                             | 3.75                              | 31.25                              |
| Hurricanes (H) (7.2)                                         | 6                             | 5                             | 4                            | 4                              | 0                                 | 4                                  |
| Hurricane Days (HD) (27.0)                                   | 20                            | 15                            | 12                           | 12                             | 0                                 | 12                                 |
| Major Hurricanes (MH) (3.2)                                  | 2                             | 2                             | 1                            | 1                              | 0                                 | 1                                  |
| Major Hurricane Days (MHD) (7.4)                             | 5                             | 4                             | 2                            | 2                              | 0                                 | 2                                  |
| Accumulated Cyclone Energy (ACE) (123)                       | 90                            | 70                            | 50                           | 50                             | 3                                 | 47                                 |
| ACE West of 60°W (73)                                        | 50                            | 35                            | 25                           | 25                             | 3                                 | 22                                 |
| Net Tropical Cyclone Activity (NTC) (135%)                   | 100                           | 80                            | 60                           | 60                             | 5                                 | 55                                 |

### PROBABILITIES FOR AT LEAST ONE MAJOR (CATEGORY 3-4-5) HURRICANE LANDFALL ON EACH OF THE FOLLOWING COASTAL AREAS (AFTER 4 AUGUST):

- 1) Entire continental U.S. coastline – 16% (full season average from 1880–2020 is 43%)
- 2) U.S. East Coast Including Peninsula Florida (south and east of Cedar Key, Florida) – 7% (full season average from 1880–2020 is 21%)
- 3) Gulf Coast from the Florida Panhandle (west and north of Cedar Key, Florida) westward to Brownsville – 9% (full season average from 1880–2020 is 27%)

### PROBABILITY FOR AT LEAST ONE MAJOR (CATEGORY 3-4-5) HURRICANE TRACKING THROUGH THE CARIBBEAN (10–20°N, 88–60°W) (AFTER 4 AUGUST):

- 1) 17% (full season average from 1880–2020 is 47%)

## ABSTRACT

Information obtained through July indicates that the 2026 Atlantic hurricane season should have activity well below the 1991–2020 average. We estimate that 2026 will have 9 named storms (average is 14.4), 35 named storm days (average is 69.4), 4 hurricanes (average is 7.2), 12 hurricane days (average is 27.0), 1 major (Category 3–4–5) hurricane (average is 3.2) and 2 major hurricane days (average is 7.4). These forecast numbers include the two tropical storms that have occurred already in 2026: Arthur and Bertha. The probability of U.S. and Caribbean major hurricane landfall is estimated to be well below its long-period average. We predict Atlantic basin Accumulated Cyclone Energy (ACE) and Net Tropical Cyclone (NTC) activity in 2026 to be ~40–45 percent of their long-term averages.

Coastal residents are reminded that it only takes one hurricane making landfall to make it an active season for them. Thorough preparations should be made for every season, regardless of how much activity is predicted.

This forecast is based on two August statistical prediction schemes that were developed using ~40 years of past data. Analog predictors are utilized as well. We are also including statistical/dynamical models based on 25–40 years of past data from the European Centre for Medium Range Weather Forecasts, the UK Met Office, the Japan Meteorological Agency and the Centro Euro-Mediterraneo sui Cambiamenti Climatici model as four additional forecast guidance tools. In addition, for the first time this year, we are incorporating forecast output from the Ai2 Climate Emulator – a machine learning-based atmospheric model. All of this model guidance points towards a below-normal season.

The tropical Pacific is currently characterized by moderate/strong El Niño conditions, with all signs pointing towards a continuing powerful El Niño for the next few months. Sea surface temperatures across the tropical Atlantic are slightly above their long-term averages. We currently anticipate that El Niño will dominate the tropical circulation during the peak of the 2026 Atlantic hurricane season, driving well above-normal levels of vertical wind shear across the tropical Atlantic and Caribbean.

The early August forecast has good long-term skill when evaluated in hindcast mode. The skill of CSU’s forecast updates increases as the peak of the Atlantic hurricane season approaches. We have high confidence in this year’s seasonal forecast given the strength of the El Niño.

Starting today and released every two weeks following (e.g., 5 August, 19 August, 2 September, etc.), we will issue two-week forecasts for Atlantic TC activity during the peak of the Atlantic hurricane season from August–October.

## **Why issue extended-range forecasts for seasonal hurricane activity?**

We are frequently asked this question. Our answer is that it is possible to say something about the probability of the coming year's hurricane activity which is superior to climatology. The Atlantic basin has the largest year-to-year variability of any of the global tropical cyclone basins. People are curious to know how active the upcoming season is likely to be, particularly if you can show hindcast skill improvement over climatology for many past years.

Everyone should realize that it is impossible to precisely predict this season's hurricane activity in early August. There is, however, much curiosity as to how global ocean and atmosphere features are presently arranged with respect to the probability of an active or inactive hurricane season for the coming year. Our early August statistical and statistical/dynamical hybrid models show strong evidence on ~25–40 years of data that significant improvement over a climatological forecast can be attained. We would never issue a seasonal hurricane forecast unless we had models developed over a long hindcast period which showed skill. We also include probabilities of exceedance to provide a visualization of the uncertainty associated with these predictions.

We issue these forecasts to satisfy the curiosity of the public and to bring attention to the hurricane problem. There is a general interest in knowing what the odds are for an active or inactive season. One must remember that our forecasts are based on the premise that those global oceanic and atmospheric conditions which preceded comparatively active or inactive hurricane seasons in the past provide meaningful information about similar trends in future seasons.

It is also important that the reader appreciate that these seasonal forecasts are based on statistical and dynamical models which will fail in some years. Moreover, these forecasts do not specifically predict where within the Atlantic basin these storms will strike. The probability of landfall for any one location along the coast is very low and reflects the fact that, in any one season, most U.S. coastal areas will not feel the effects of a hurricane no matter how active the individual season is.

### Acknowledgment

These seasonal forecasts were developed by the late Dr. William Gray, who was lead author on these predictions for over 20 years and continued as a co-author until his death in 2016. In addition to pioneering seasonal Atlantic hurricane prediction, he conducted groundbreaking research on a wide variety of other topics including hurricane genesis, hurricane structure and cumulus convection that are discussed in a [paper](#) highlighting his research legacy. His investments in both time and energy on these forecasts cannot be acknowledged enough.

We are grateful for support from Commodity Weather Group, Gallagher Re, the Insurance Information Institute, Ironshore Insurance, and IAA. We acknowledge a grant from the G. Unger Vetlesen Foundation for additional financial support.

Colorado State University's seasonal hurricane forecasts have benefited greatly from several individuals that were former graduate students of William Gray. Among these former project members are Chris Landsea, John Knaff and Eric Blake. We also would like to thank Jhordanne Jones and Alex DesRosiers, Ph.D. graduates from Michael Bell's research group, for model development and forecast assistance over the past several years. Thanks also extend to current members of the Tropical Cyclones, Radar, Atmospheric Modeling, and Software (TC-RAMS) team who have provided valuable comments and feedback throughout the forecast preparation process. These members include: Tyler Barbero, Lauren Beard, Deztynee Bryan, Ming-Han Chung, Jen DeHart, Chandler Jenkins, Jinhui Ju and Meghan Stell.

We thank Louis-Philippe Caron and the data team at the Barcelona Supercomputing Centre for providing data and insight on the statistical/dynamical models. We have also benefited from meteorological discussions with Louis-Philippe Caron, Dan Chavas, Jason Dunion, Michael Lowry, Brian McNoldy, Paul Roundy, Mike Ventrice and Peng Xian over the past few years.

## DEFINITIONS AND ACRONYMS

Accumulated Cyclone Energy (ACE) - A measure of a named storm's potential for wind destruction defined as the sum of the square of a named storm's maximum wind speed (in  $10^4$  knots<sup>2</sup>) for each 6-hour period of its existence. The 1991–2020 average value of this parameter is 123 for the Atlantic basin.

Atlantic Multi-Decadal Oscillation (AMO) – A mode of natural variability that occurs in the North Atlantic Ocean and evidencing itself in fluctuations in sea surface temperature and sea level pressure fields. The AMO is likely related to fluctuations in the strength of the oceanic thermohaline circulation. Although several definitions of the AMO are currently used in the literature, we define the AMO based on North Atlantic sea surface temperatures from 50–60°N, 50–10°W and sea level pressure from 0–50°N, 70–10°W.

Atlantic Basin – The area including the entire North Atlantic Ocean, the Caribbean Sea, and the Gulf of Mexico.

El Niño – A 12-18 month period during which anomalously warm sea surface temperatures occur in the eastern half of the equatorial Pacific. Moderate or strong El Niño events occur irregularly, about once every 3–7 years on average.

Hurricane (H) - A tropical cyclone with sustained low-level winds of 74 miles per hour ( $33 \text{ ms}^{-1}$  or 64 knots) or greater.

Hurricane Day (HD) - A measure of hurricane activity, one unit of which occurs as four 6-hour periods during which a tropical cyclone is observed or is estimated to have hurricane-force winds.

Indian Ocean Dipole (IOD) - An irregular oscillation of sea surface temperatures between the western and eastern tropical Indian Ocean. A positive phase of the IOD occurs when the western Indian Ocean is anomalously warm compared with the eastern Indian Ocean.

Madden Julian Oscillation (MJO) – A globally propagating mode of tropical atmospheric intra-seasonal variability. The wave tends to propagate eastward at approximately  $5 \text{ ms}^{-1}$ , circling the globe in roughly 30-60 days.

Main Development Region (MDR) – The region in the tropical Atlantic and Caribbean where most tropical cyclones that become major hurricanes form. We define the MDR to span 10–20°N, 85–20°W.

Major Hurricane (MH) - A hurricane which reaches a sustained low-level wind of at least 111 mph (96 knots or  $50 \text{ ms}^{-1}$ ) at some point in its lifetime. This constitutes a category 3 or higher on the Saffir/Simpson scale.

Major Hurricane Day (MHD) - Four 6-hour periods during which a hurricane has an intensity of Saffir/Simpson category 3 or higher.

Named Storm (NS) - A hurricane, a tropical storm or a sub-tropical storm.

Named Storm Day (NSD) - As in HD but for four 6-hour periods during which a tropical or sub-tropical cyclone is observed (or is estimated) to have attained tropical storm-force winds.

Net Tropical Cyclone (NTC) Activity – Average seasonal percentage mean of NS, NSD, H, HD, MH, MHD. Gives overall indication of Atlantic basin seasonal hurricane activity. The 1991–2020 average value of this parameter is 135.

Oceanic Nino Index (ONI) – Three-month running mean of SST anomalies in the Nino 3.4 region (5°S–5°N, 170–120°W) based on centered 30-year base periods.

Relative Oceanic Nino Index (RONI) – Three-month running mean of SST anomalies in the Nino 3.4 region (5°S–5°N, 170–120°W) minus tropically-averaged (20°S–20°N) SST anomalies multiplied by a scaling factor.

Saffir/Simpson Hurricane Wind Scale – A measurement scale ranging from 1 to 5 of hurricane wind intensity. One is a weak hurricane; whereas, five is the most intense hurricane.

Southern Oscillation Index (SOI) – A normalized measure of the surface pressure difference between Tahiti and Darwin. Low values typically indicate El Niño conditions.

Standard Deviation (SD) – A measure used to quantify the variation in a dataset.

Sea Surface Temperature Anomaly (SSTA) – Observed sea surface temperature differenced from a long-period average, typically 1991–2020.

Thermohaline Circulation – A large-scale circulation in the Atlantic Ocean that is driven by fluctuations in salinity and temperature. When the thermohaline circulation is stronger than normal, the AMO tends to be in its warm (or positive) phase, and more Atlantic hurricanes typically form.

Tropical Cyclone (TC) - A large-scale circular flow occurring within the tropics and subtropics which has its strongest winds at low levels; including hurricanes, tropical storms and other weaker rotating vortices.

Tropical Storm (TS) - A tropical cyclone with maximum sustained winds between 39 mph ( $18 \text{ ms}^{-1}$  or 34 knots) and 73 mph ( $32 \text{ ms}^{-1}$  or 63 knots).

Vertical Wind Shear – The difference in horizontal wind between 200 hPa (approximately 40000 feet or 12 km) and 850 hPa (approximately 5000 feet or 1.6 km).

1 knot = 1.15 miles per hour = 0.515 meters per second

# 1 Introduction

This is the 43rd year in which the TC-RAMS team has made forecasts of the upcoming season’s Atlantic basin hurricane activity. Our research team has shown that a sizable portion of the year-to-year variability of Atlantic tropical cyclone (TC) activity can be forecast with skill exceeding climatology. Several components are used to produce our August forecast. These components include two statistical regression models, a combined statistical/dynamical model from four different modeling centers, a machine learning-based model, a selection of analog seasons, and lastly, qualitative adjustments to accommodate additional processes which may not be explicitly represented by these analyses. The statistical/dynamical models are from the European Centre for Medium-Range Weather Forecasts (ECMWF), the UK Met Office, the Japan Meteorological Agency and the Centro Euro-Mediterraneo sui Cambiamenti Climatici (CMCC). All of these models show skill at predicting TC activity based on ~25–40 years of historical data. For the first time this year, we are also incorporating forecasts from the Ai2 Climate Emulator (ACE2). These evolving forecast techniques are based on a variety of climate-related global and regional predictors previously shown to be related to the forthcoming seasonal Atlantic basin TC activity and landfall probability. We believe that seasonal forecasts must be based on methods that show significant hindcast skill in application to long periods of prior data. It is only through hindcast skill that one can demonstrate that seasonal forecast skill is possible. This is a valid methodology provided that the atmosphere continues to behave in the future as it has in the past.

The best predictors do not necessarily have the best individual correlations with hurricane activity. The best forecast parameters are those that explain the portion of the variance of seasonal hurricane activity that are not associated with the other forecast variables. It is possible for an important hurricane forecast parameter to show little direct relationship to a predictand by itself but to have an important influence when included with a set of 2–3 other predictors.

## 2.1 August Statistical Forecast Scheme using July Data

We developed a 1 August statistical seasonal forecast scheme for the prediction of Accumulated Cyclone Energy (ACE) that has been utilized operationally since 2012. The model was updated in 2023 to use ERA5 data.

The pool of three predictors for the early August statistical forecast scheme is given in Table 1. The location of each of these predictors are shown in Figure 1. Skillful forecasts can be issued for post-31 July ACE based upon cross-validated hindcasts from 1979–2025. When these three predictors are combined, they predict an ACE that correlates at 0.80 with observed ACE using cross-validated hindcasts from 1979–2025 (Figure 2). This model favors below-average Atlantic hurricane activity in 2026.

Table 1: Listing of 1 August 2026 predictors for this year's hurricane activity. A plus (+) means that positive deviations of the parameter indicate increased hurricane activity this year, and a minus (-) means that positive deviations of the parameter indicate decreased hurricane activity this year.

| Predictor                                  | Values for 2026 Forecast | Effect on 2026 Hurricane Season |
|--------------------------------------------|--------------------------|---------------------------------|
| 1) July 10 m U (7.5-17.5°N, 85-60°W) (+)   | -1.3 SD                  | Suppress                        |
| 2) July SST (20-40°N, 35-15°W) (+)         | +1.5 SD                  | Enhance                         |
| 3) July 200 hPa U (10-20°N, 30°W-30°E) (-) | +2.5 SD                  | Suppress                        |

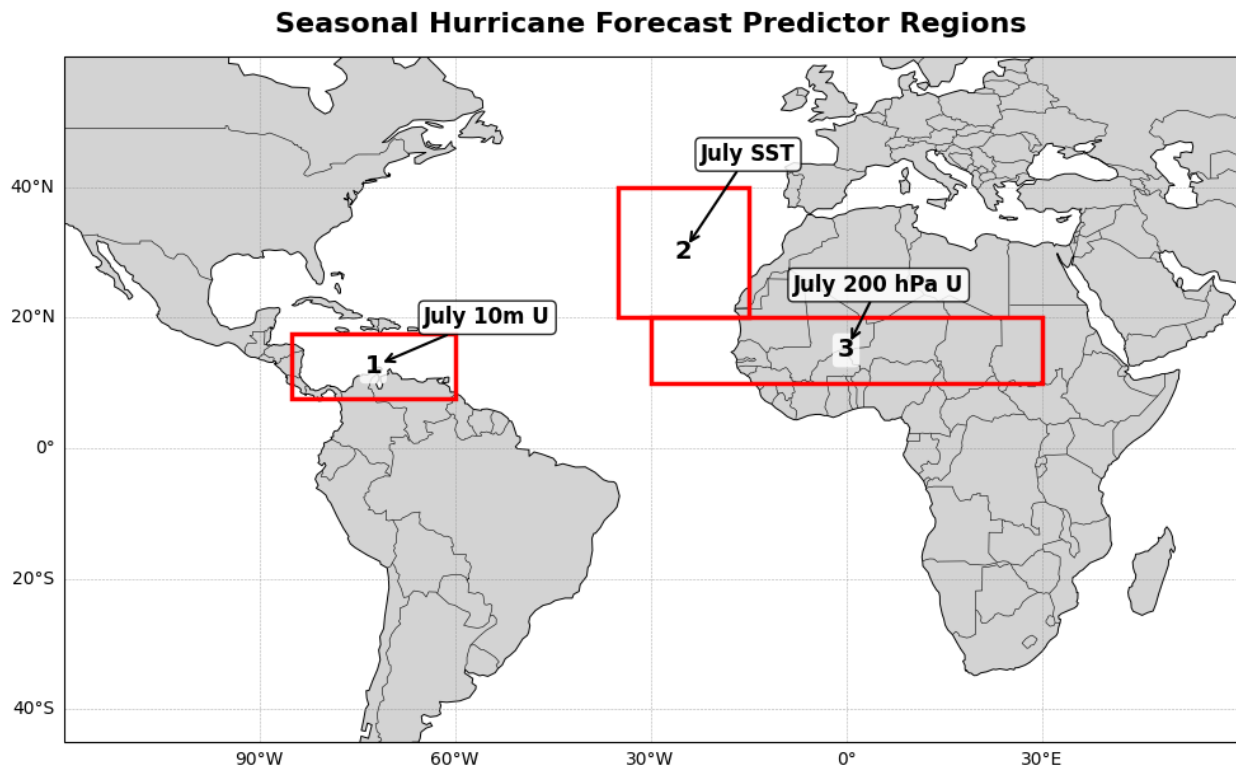


Figure 1: Location of predictors for the post-31 July forecast for the 2026 hurricane season from the July-averaged statistical model.

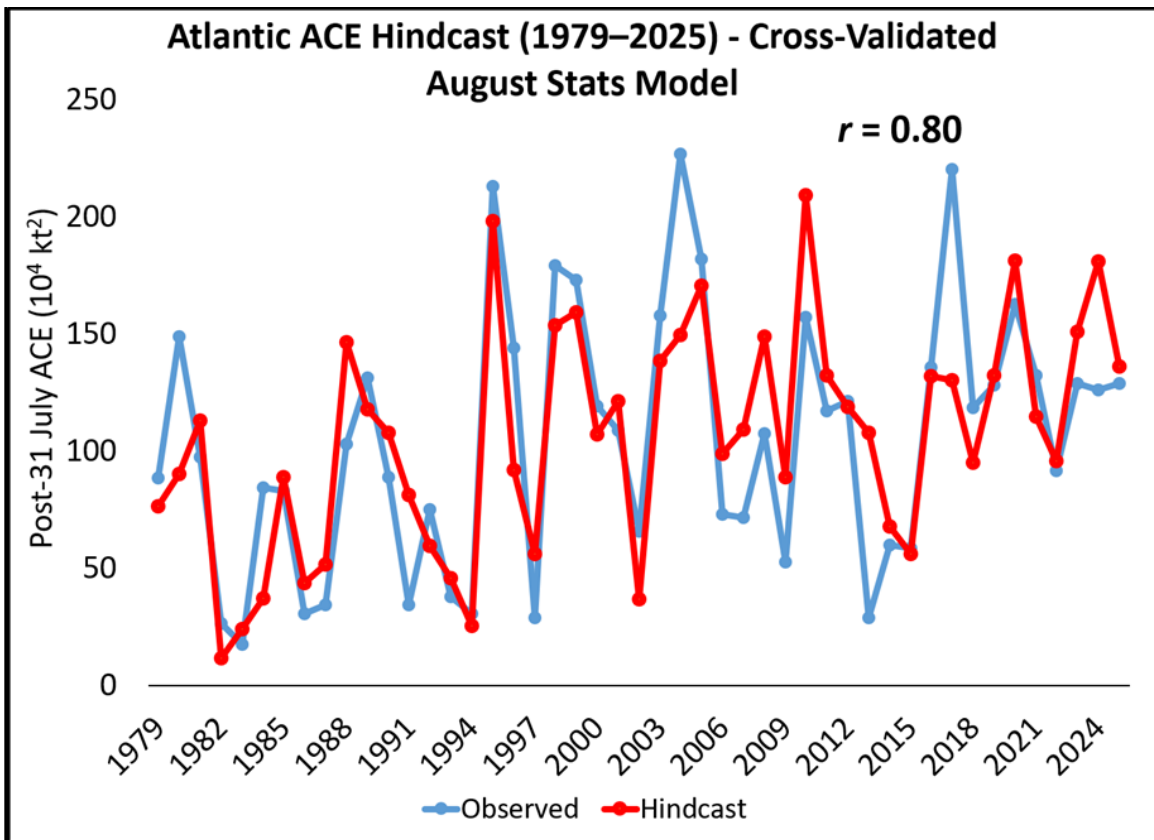


Figure 2: Observed versus hindcast values of post-31 July ACE for 1979–2025 using our statistical scheme that uses July averages.

Table 2 shows our forecast for the 2026 hurricane season from the July-averaged statistical model and the comparison of this forecast with the 1991–2020 average. The statistical forecast is calling for a below-average remainder of the season.

Table 2: Post-31 July statistical forecast for 2026 from the July-averaged statistical model.

| Predictands and Climatology (1991–2020<br>Post-31 July Average) | Post-31 July<br>Statistical<br>Forecast | Full Season Statistical<br>Forecast (Activity Thru 31<br>July Added In) |
|-----------------------------------------------------------------|-----------------------------------------|-------------------------------------------------------------------------|
| Named Storms (NS) – 11.6                                        | 10.7                                    | 12.7                                                                    |
| Named Storm Days (NSD) – 61.3                                   | 43.6                                    | 47.4                                                                    |
| Hurricanes (H) – 6.5                                            | 4.5                                     | 4.5                                                                     |
| Hurricane Days (HD) – 25.6                                      | 12.9                                    | 12.9                                                                    |
| Major Hurricanes (MH) – 3.1                                     | 1.6                                     | 1.6                                                                     |
| Major Hurricane Days (MHD) – 7.1                                | 2.3                                     | 2.3                                                                     |
| Accumulated Cyclone Energy (ACE) – 113                          | 61                                      | 64                                                                      |
| Net Tropical Cyclone Activity (NTC) – 123                       | 74                                      | 79                                                                      |

### 2.1a Physical Associations among Predictors Listed in Table 1

The locations and brief descriptions of the three predictors for the July-averaged August statistical forecast are now discussed. It should be noted that all forecast parameters correlate significantly with physical features during August through October that are known to be favorable for elevated levels of TC activity. For each of these predictors, we display a three-panel figure showing correlations since 1979 between values of each predictor and August–October values of SST, sea level pressure (SLP), and 200 minus 850 hPa vertical wind shear.

#### Predictor 1. July 10 meter U in the Caribbean (+)

(7.5–17.5°N, 85–60°W)

Low-level trade wind flow has been utilized as a predictor in seasonal forecasting systems for the Atlantic basin (Saunders and Lea 2008). When the trades are weaker-than-normal, SSTs across the tropical Atlantic tend to be elevated, and consequently a larger-than-normal Atlantic Warm Pool (AWP) is typically observed (Wang and Lee 2007) (Figure 3). A larger AWP also correlates with reduced vertical shear across the tropical Atlantic. Weaker trade winds are typically associated with higher pressure in the tropical eastern Pacific (a La Niña signal) and lower pressure in the Caribbean and tropical Atlantic. Both of these conditions generally occur when active hurricane seasons are observed. Predictor 1 also has a strong negative correlation with August–October-vertical wind shear.

#### Predictor 2. July SST in the Northeastern Subtropical Atlantic (+)

(20°–40°N, 35–15°W)

A similar predictor was utilized in earlier August seasonal forecast models (Klotzbach 2007, Klotzbach 2011). Anomalously warm SSTs in the subtropical North Atlantic are associated with a positive phase of the Atlantic Meridional Mode (AMM), a northward-shifted Intertropical Convergence Zone, and consequently, reduced trade wind strength (Kossin and Vimont 2007). Weaker trade winds are associated with less surface evaporative cooling and less mixing and upwelling. This results in warmer tropical Atlantic SSTs during the August–October period (Figure 4).

Predictor 3. July 200 hPa U over the Tropical Eastern Atlantic and Northern Tropical Africa (-)

(10–20°N, 30°W–30°E)

Anomalous easterly flow at upper levels over the eastern Atlantic and northern tropical Africa provides an environment that is more favorable for easterly wave development into TCs. This anomalous easterly flow tends to persist through August–October, which reduces shear over the Main Development Region (MDR). This predictor also correlates with sea level pressure (SLP) and SST anomalies over the tropical eastern Pacific that are typically associated with cool ENSO conditions (Figure 5).

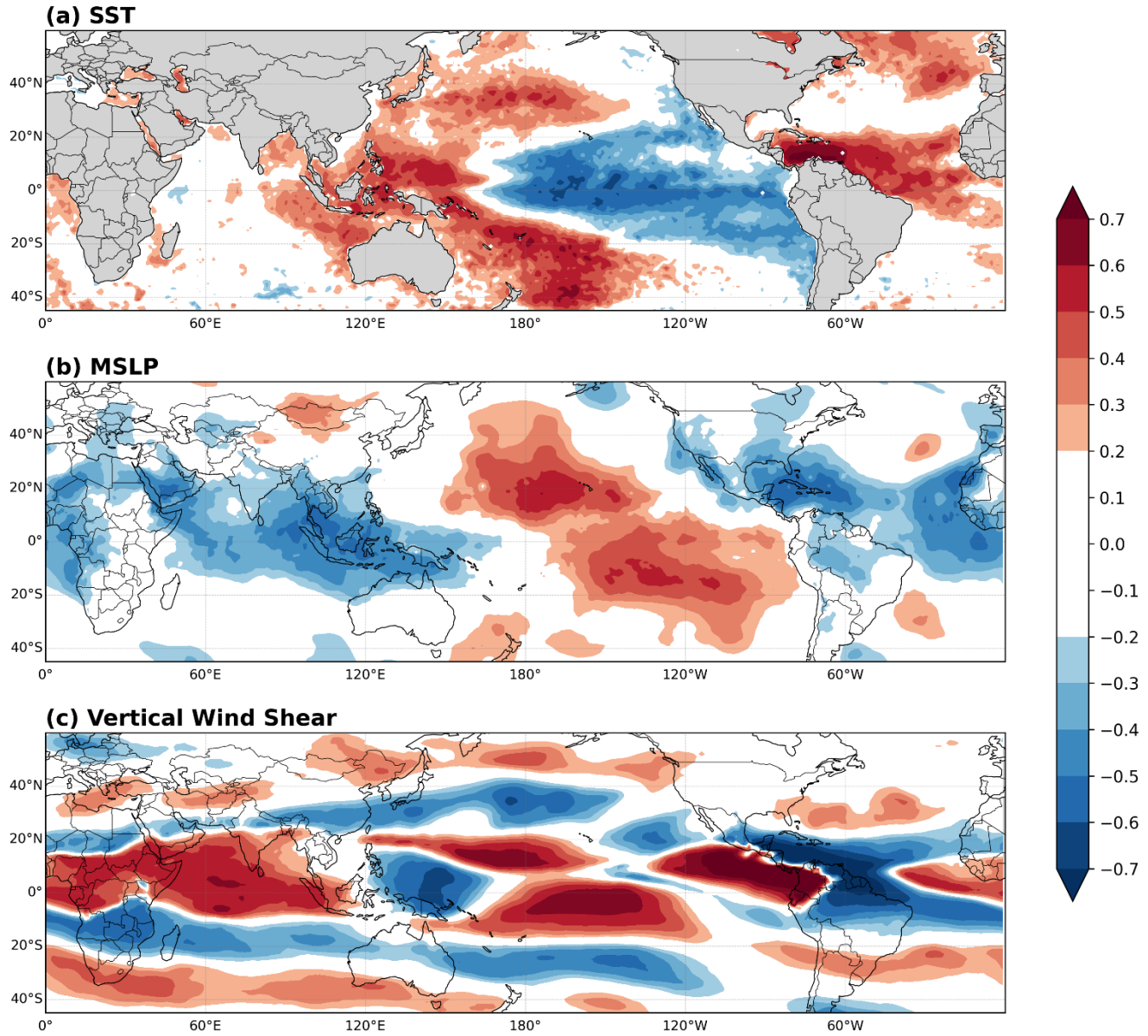


Figure 3: Correlations between July 10 meter U in the Caribbean and (a) August–October sea surface temperature, (b) August–October sea level pressure, and (c) August–October 200 minus 850 hPa vertical wind shear.

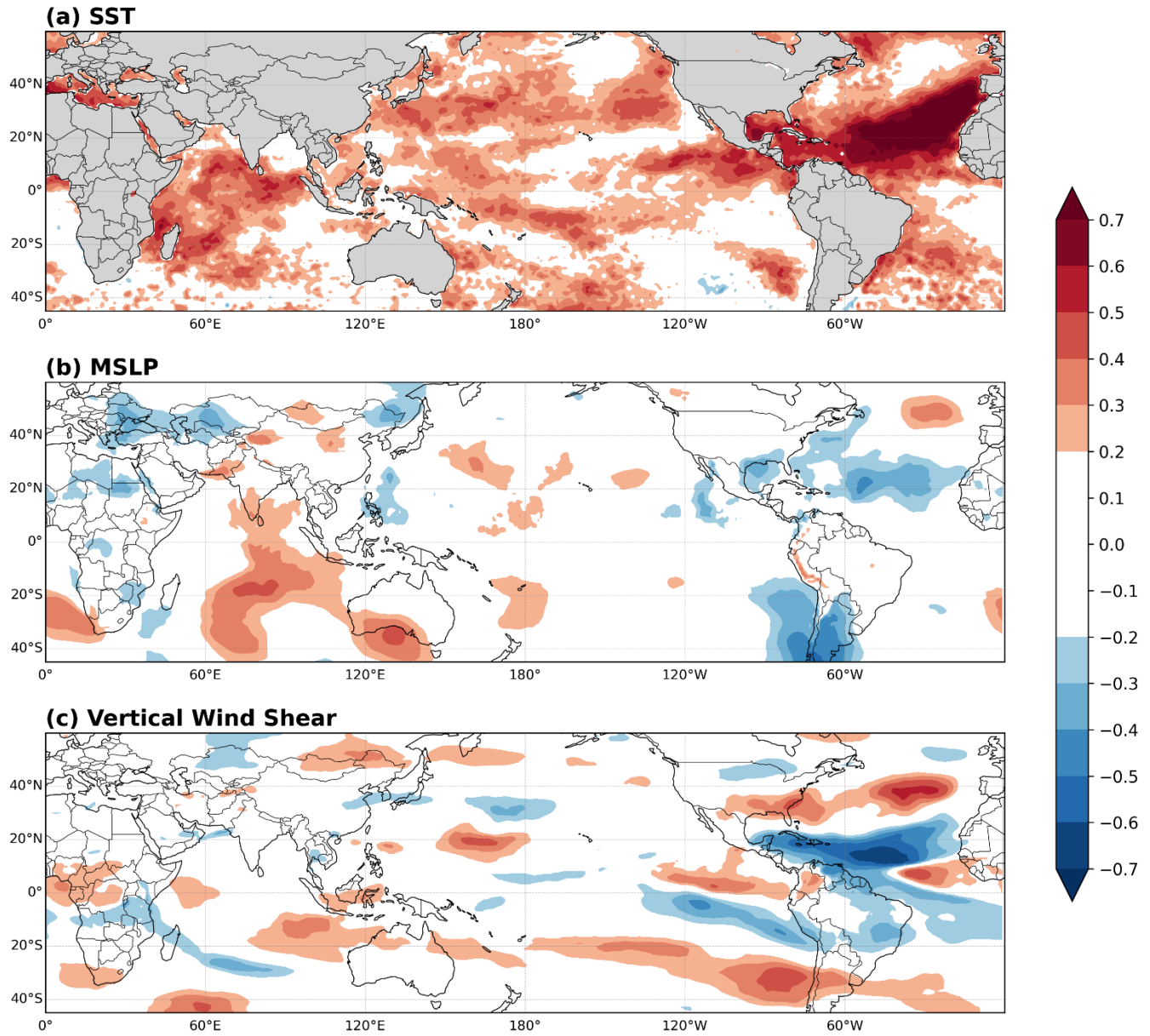


Figure 4: As in Figure 3 but for July sea surface temperature in the subtropical northeastern Atlantic.

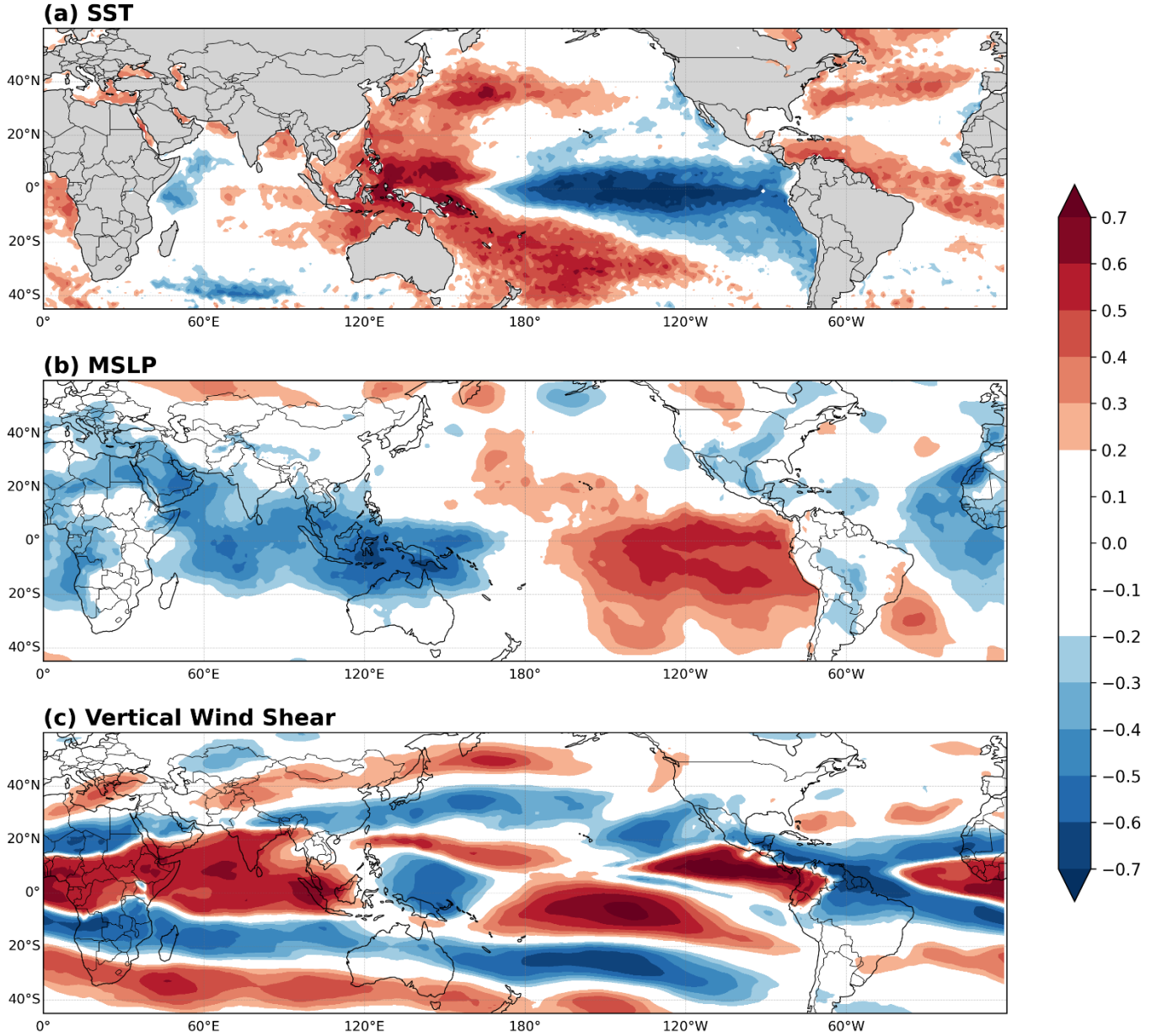


Figure 5: As in Figure 3 but for July 200 hPa zonal wind over tropical north Africa. Predictor values have been multiplied by -1 so that the signs of correlations match up with those in Figures 3 and 4.

## 2.2 August Statistical Forecast Scheme using 50-Day Averages

We now include a second August statistical seasonal forecast scheme that uses 50-day averages, to complement the scheme that uses July-only data. The reason for using the longer averages is to reduce the impact of sub-seasonal variability, such as the Madden-Julian oscillation, which can impart atmospheric signals on shorter timescales that may not be representative of the longer-term signal that is critical for seasonal forecasting. This model uses similar predictors to what is used with our model using July averages, with slight tweaks to the predictor boundaries to capture where these predictors

showed higher skill in mid- to late June. Given that the boundaries and physical reasonings are similar between our 50-day-averaged and July-averaged statistical model, we do not include a separate discussion of the physical reasoning behind each of the three predictors selected for the 50-day-average model.

This statistical model also uses ERA5 data. The model was developed on data from 1979–2021 and has been issued operationally since 2022.

The pool of three predictors for the 50-day-averaged early August statistical forecast scheme is defined in Table 3. The location of each of these predictors is shown in Figure 6. Skillful forecasts can be issued for post-31 July ACE based upon cross-validated hindcasts from 1979–2021 and real-time forecasts from 2022–2025. When these three predictors are combined, they correlate at 0.77 with observed ACE using cross-validated hindcasts from 1979–2025 (Figure 7). This model also calls for a below-average remainder of the Atlantic hurricane season.

Table 3: Listing of 50-day-averaged statistical model values for the August 2026 hurricane forecast. A plus (+) means that positive deviations of the parameter indicate increased hurricane activity this year, and a minus (-) means that positive deviations of the parameter indicate decreased hurricane activity this year.

| Predictor                                                 | Values for<br>2026 Forecast | Effect on 2026<br>Hurricane Season |
|-----------------------------------------------------------|-----------------------------|------------------------------------|
| 1) June 12 – July 31 10 m U (5-20°N, 90-60°W) (+)         | -1.9 SD                     | Suppress                           |
| 2) June 12 – July 31 SST (25-50°N, 30-10°W) (+)           | +1.5 SD                     | Enhance                            |
| 3) June 12 – July 31 200 hPa U (15°S-15°N, 20°W-40°E) (-) | +2.2 SD                     | Suppress                           |

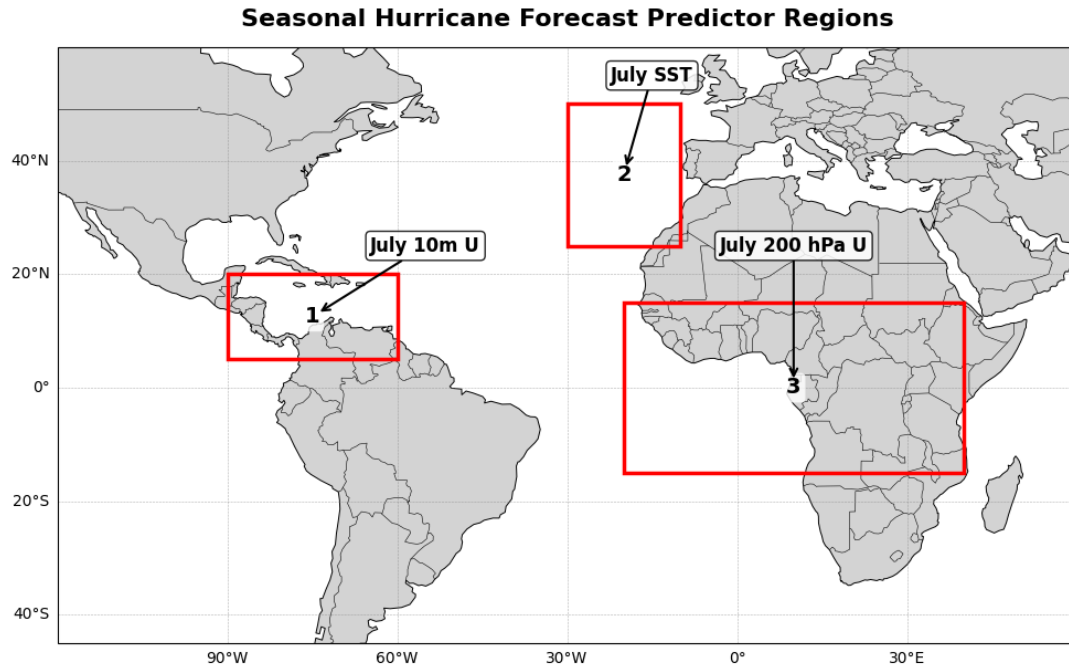


Figure 6: Location of predictors for the post-31 July forecast for the 2026 hurricane season from the 50-day-averaged (e.g., June 12 – July 31) statistical model.

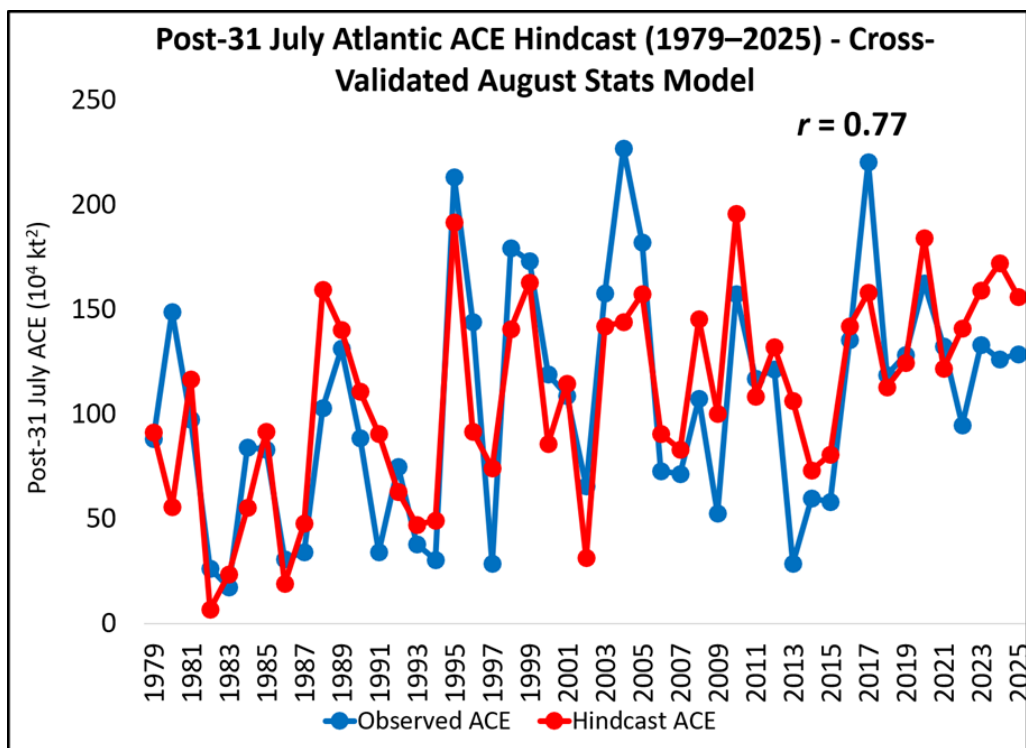


Figure 7: Observed versus hindcast values of post-31 July ACE for 1979–2025 using the 50-day-average statistical model.

Table 4 shows our forecast for the 2026 hurricane season from the statistical model using 50-day averages and the comparison of this forecast with the 1991–2020 average.

Table 4: Post-31 July statistical forecast for 2026 from the 50-day averaged statistical model.

| Predictands and Climatology (1991-2020<br>Post-31 July Average) | Post-31 July<br>Statistical<br>Forecast | Full Season Statistical<br>Forecast (Activity Thru 31<br>July Added In) |
|-----------------------------------------------------------------|-----------------------------------------|-------------------------------------------------------------------------|
| Named Storms (NS) – 11.6                                        | 11.0                                    | 13.0                                                                    |
| Named Storm Days (NSD) – 61.3                                   | 46.1                                    | 49.8                                                                    |
| Hurricanes (H) – 6.5                                            | 4.7                                     | 4.7                                                                     |
| Hurricane Days (HD) – 25.6                                      | 14.3                                    | 14.3                                                                    |
| Major Hurricanes (MH) – 3.1                                     | 1.7                                     | 1.7                                                                     |
| Major Hurricane Days (MHD) – 7.1                                | 2.8                                     | 2.8                                                                     |
| Accumulated Cyclone Energy (ACE) – 113                          | 67                                      | 70                                                                      |
| Net Tropical Cyclone Activity (NTC) – 123                       | 80                                      | 85                                                                      |

### 2.3 Statistical/Dynamical Forecast Schemes

We have used statistical/dynamical hybrid forecast models at CSU since 2019. These models, developed in partnership with Louis-Philippe Caron and the data team at the Barcelona Supercomputing Centre, originally used output from the ECMWF SEAS5 model to forecast the input to our early August statistical forecast model ([Klotzbach et al. 2020](#)). For this forecast, we are incorporating forecasts of August–September western Atlantic (10°N–20°N, 85°W–50°W) 200 minus 850 hPa vertical wind shear from four different models initialized on 1 July, namely, ECMWF, UK Met, JMA and CMCC. We then regress each model’s forecasts of western Atlantic vertical wind shear to forecast ACE for the 2026 season. All four models have hindcast data available from 1993–2016, so the regression models are developed on data during this time period. We use a log linear regression approach and set an ACE floor of 15 to keep results within the bounds of what has been observed for ACE since 1950. The 17 ACE observed in 1983 is the lowest seasonal ACE since 1950.

Figure 8 displays the location of the vertical wind shear predictor, while Table 5 displays the various statistical/dynamical model forecasts of western Atlantic vertical wind shear. Table 6 displays the seasonal TC forecast output for each of the statistical/dynamical models. All four of these models are forecasting a well below-normal Atlantic hurricane season given the extremely high western Atlantic vertical wind shear that is being forecast. These high vertical wind shear forecasts are due to the powerful El Niño that is being forecast by all of these models.

Figure 9 displays the model hindcasts of western Atlantic vertical wind shear from 1993–2016, while Figure 10 displays the model hindcasts of ACE using the log linear regression relationship between western Atlantic vertical wind shear and Atlantic ACE. All models show robust skill at forecasting western Atlantic vertical wind shear as well as Atlantic ACE.

Figure 11 displays the ECMWF ensemble average August–September SST forecast for 45°S–60°N, while Figure 12 displays the ECMWF ensemble average September 200 minus 850 hPa wind shear forecast for the North Atlantic. As would be expected given the very strong El Niño that is being forecast by the ECMWF ensemble, September vertical wind shear across the MDR is forecast to be the highest on record in the ECMWF hindcast/forecast period since 1981.

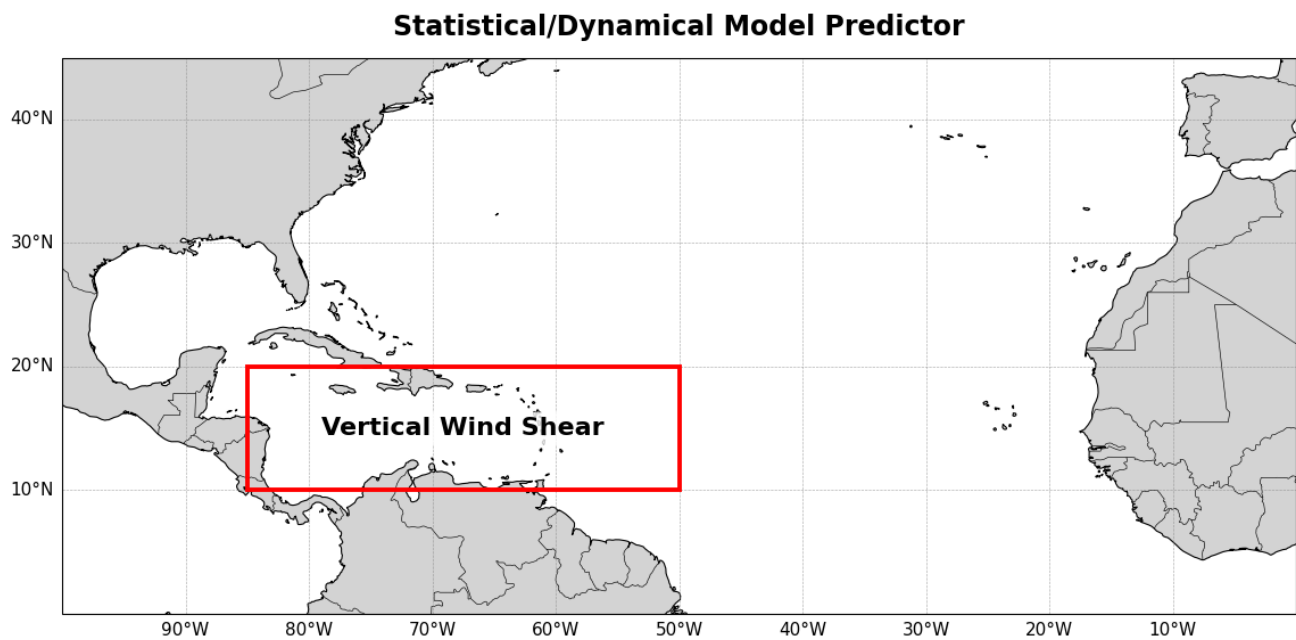


Figure 8: Location of our western Atlantic vertical wind shear predictor for the statistical/dynamical model.

Table 5: Listing of predictions of August–September western Atlantic vertical wind shear from our statistical/dynamical model output, initialized on 1 July. A plus (+) means that positive deviations of the parameter are associated with increased hurricane activity, while a minus (-) means that negative deviations of the parameter are associated with increased hurricane activity. Deviations are calculated relative to the model’s 1993–2016 hindcast period.

| Predictor                             | ECMWF<br>Forecast | UK Met<br>Forecast | JMA<br>Forecast | CMCC<br>Forecast |
|---------------------------------------|-------------------|--------------------|-----------------|------------------|
| 1) Aug–Sep VWS (10–20°N, 85–50°W) (+) | +3.0 SD           | +2.0 SD            | +2.5 SD         | +2.3 SD          |

Table 6: Summary of our statistical/dynamical forecasts.

| Forecast Parameter and 1991–2020 Average<br>(in parentheses) | ECMWF<br>Scheme | Met Office<br>Scheme | JMA<br>Scheme | CMCC<br>Scheme | Adjusted Final<br>Forecast |
|--------------------------------------------------------------|-----------------|----------------------|---------------|----------------|----------------------------|
| Named Storms (14.4)                                          | 8.8             | 10.0                 | 9.4           | 9.9            | 9                          |
| Named Storm Days (69.4)                                      | 31.0            | 38.9                 | 34.7          | 38.1           | 35                         |
| Hurricanes (7.2)                                             | 3.1             | 4.0                  | 3.5           | 3.9            | 4                          |
| Hurricane Days (27.0)                                        | 6.1             | 10.4                 | 8.1           | 9.9            | 12                         |
| Major Hurricanes (3.2)                                       | 0.8             | 1.3                  | 1.0           | 1.2            | 1                          |
| Major Hurricane Days (7.4)                                   | 0.5             | 1.4                  | 1.0           | 1.2            | 2                          |
| Accumulated Cyclone Energy Index (123)                       | 31              | 50                   | 40            | 48             | 50                         |
| Net Tropical Cyclone Activity (135%)                         | 46              | 63                   | 53            | 61             | 60                         |

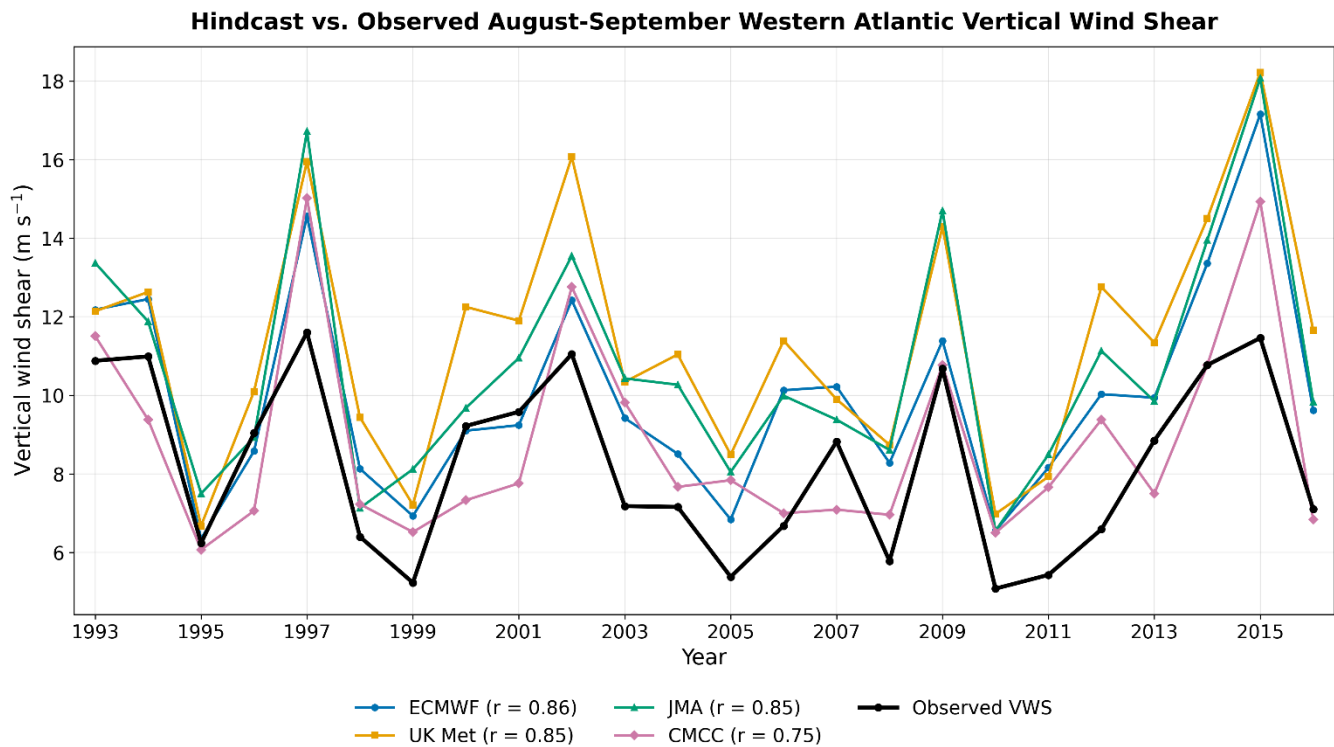


Figure 9: Observed versus hindcast values of western Atlantic vertical wind shear (VWS) from 1993–2016. Correlations between each modeling agency’s hindcasts and observed vertical wind shear is listed in the legend.

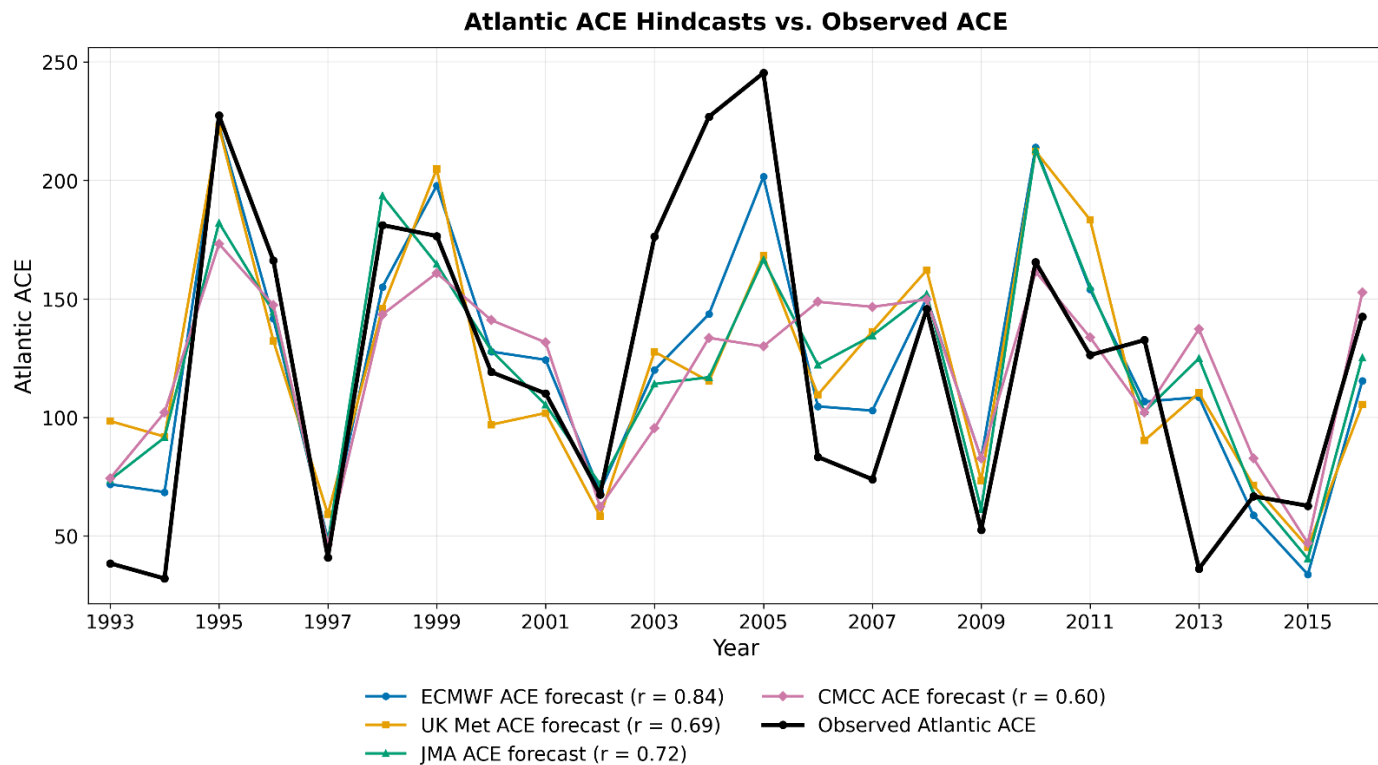


Figure 10: Observed versus statistical/dynamical hindcasts of seasonal ACE for all four statistical/dynamical models from 1993–2016. Correlations between each modeling agency’s hindcasts and observed ACE is listed in the legend.

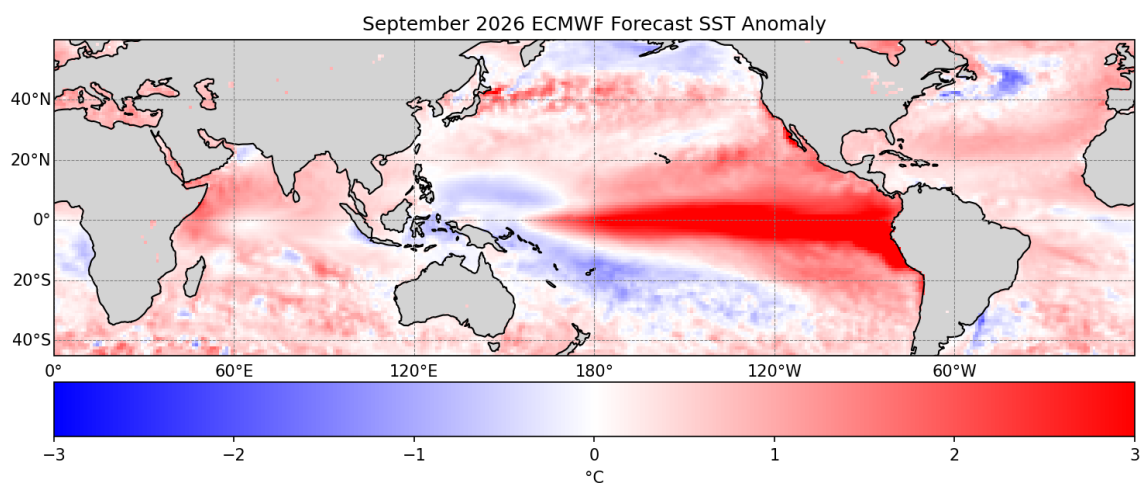


Figure 11: ECMWF ensemble average SST anomaly forecast for September 2026.

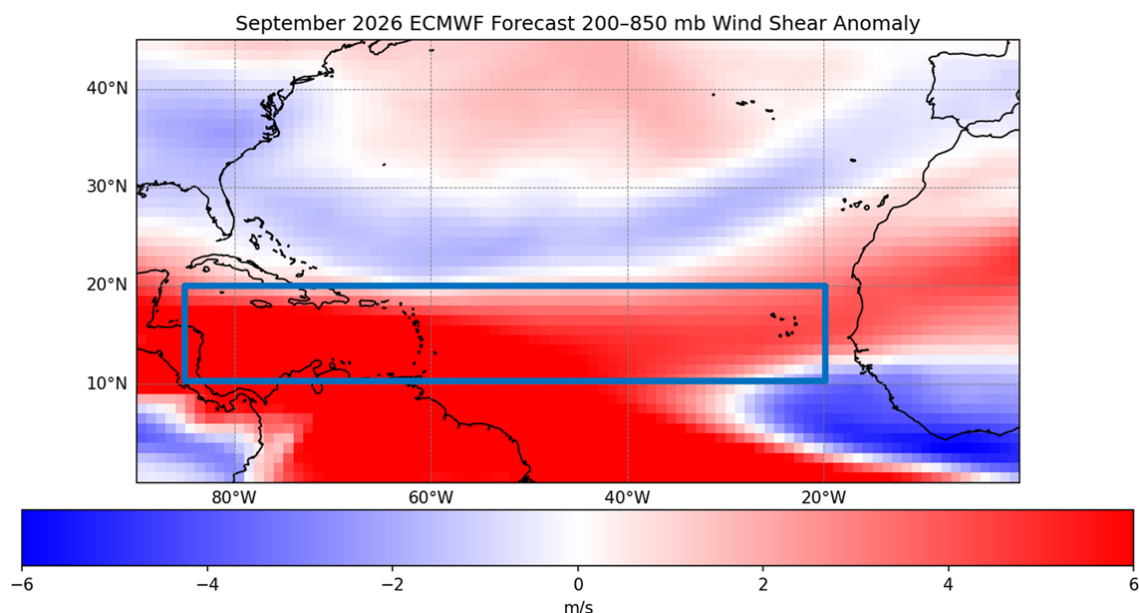


Figure 12: ECMWF ensemble average 200 minus 850 hPa anomalous wind shear forecast for September 2026. The blue rectangle denotes the MDR.

### 2.3 Machine Learning-Based Model Forecast

This year we are debuting a new analysis technique for our seasonal forecasts. We use the Ai2 Climate Emulator (ACE2) ([Watt-Meyer et al. 2025](#)) – a machine learning-based atmospheric model – as an additional guidance tool for our outlooks. Two practical benefits of using a machine learning-based model are the speed of calculations and the resulting improved ability to quantify the uncertainty in our forecast. Recent research has shown that ACE2 can accurately emulate subseasonal and seasonal variability, including the Madden-Julian Oscillation and the atmospheric response to El Niño variability, over recent decades. The version of ACE2 we use was trained with historically informed, observationally driven data from ERA5. ACE2 is able to predict the future state of the atmosphere with user-input SST and concentration of CO<sub>2</sub>. We force ACE2 with the 51 members comprising the ECMWF ensemble prediction of SSTs through September 2026. Figure 13 displays the September ensemble mean vertical wind shear from ACE2 across the tropical and subtropical Atlantic. The ACE2 model is consistent with results from ECMWF that the MDR-averaged vertical wind shear should be quite pronounced during the peak of the Atlantic hurricane season.

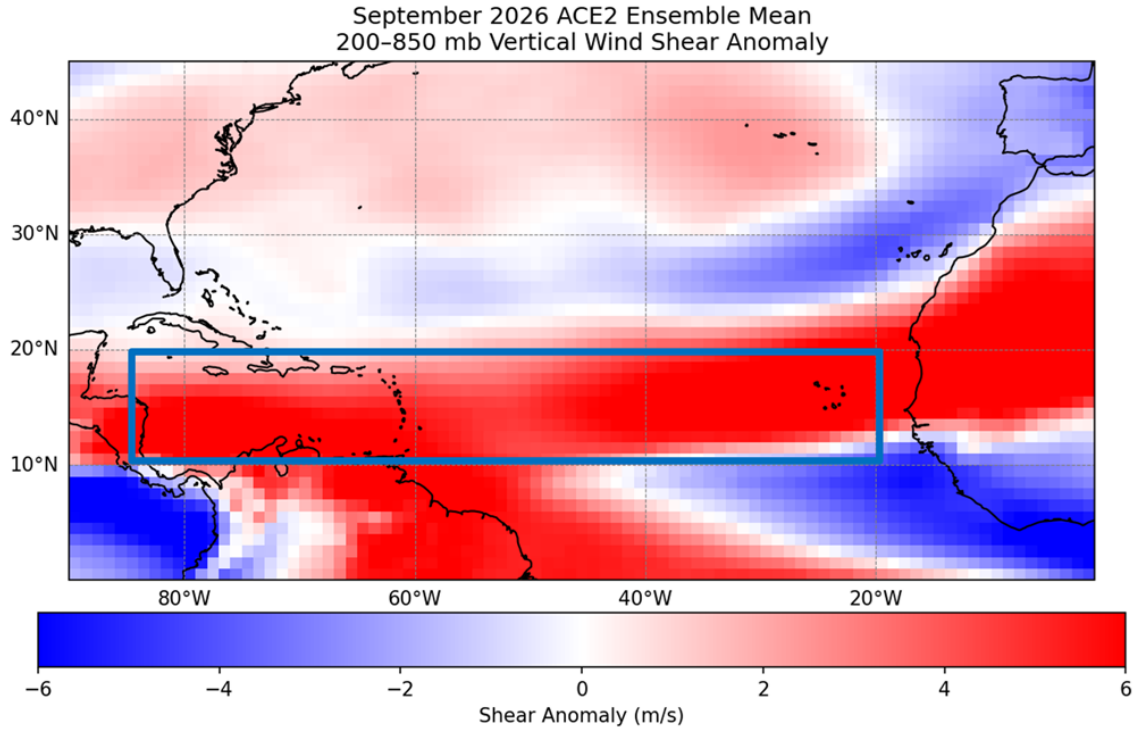


Figure 13: ACE2 ensemble average anomalous vertical wind shear (200 minus 850 hPa) forecast for September 2026. The blue rectangle denotes the MDR.

## 2.4 August Analog Forecast Scheme

Certain years in the historical record have global oceanic and atmospheric trends which are similar to 2026. These years also provide useful clues as to likely levels of activity that the forthcoming 2026 hurricane season may bring. For this early August extended range forecast, we determine which of the prior years in our database have distinct trends in key environmental conditions which are similar to current conditions and, more importantly, projected August–October 2026 conditions. We have kept the same analogs as we had with our July forecast. Table 7 lists our analog selections, while Figure 14 shows the composite August–October SST anomalies in our six analog years.

We searched for years that had moderate to strong El Niño conditions and near average tropical Atlantic SST conditions. We anticipate that the 2026 hurricane season will have activity near the average of our six analog years for most parameters.

Table 7: Analog years for 2026 with the associated hurricane activity listed for each year.

| Year                 | NS       | NSD       | H        | HD        | MH       | MHD      | ACE       | NTC       |
|----------------------|----------|-----------|----------|-----------|----------|----------|-----------|-----------|
| 1965                 | 10       | 48.25     | 4        | 19.50     | 1        | 6.25     | 86.7      | 86.4      |
| 1982                 | 6        | 18.50     | 2        | 5.75      | 1        | 1.25     | 31.5      | 37.7      |
| 1987                 | 7        | 37.25     | 3        | 5.00      | 1        | 0.50     | 34.4      | 45.6      |
| 1997                 | 8        | 30.00     | 3        | 9.50      | 1        | 2.25     | 40.9      | 53.8      |
| 2009                 | 9        | 30.00     | 3        | 12.00     | 2        | 3.50     | 52.6      | 68.6      |
| 2015                 | 11       | 43.50     | 4        | 12.00     | 2        | 4.00     | 62.7      | 81.2      |
| Average              | 8.5      | 34.6      | 3.2      | 10.6      | 1.3      | 3.0      | 51.5      | 62.2      |
| <b>2026 Forecast</b> | <b>9</b> | <b>35</b> | <b>4</b> | <b>12</b> | <b>1</b> | <b>2</b> | <b>50</b> | <b>60</b> |

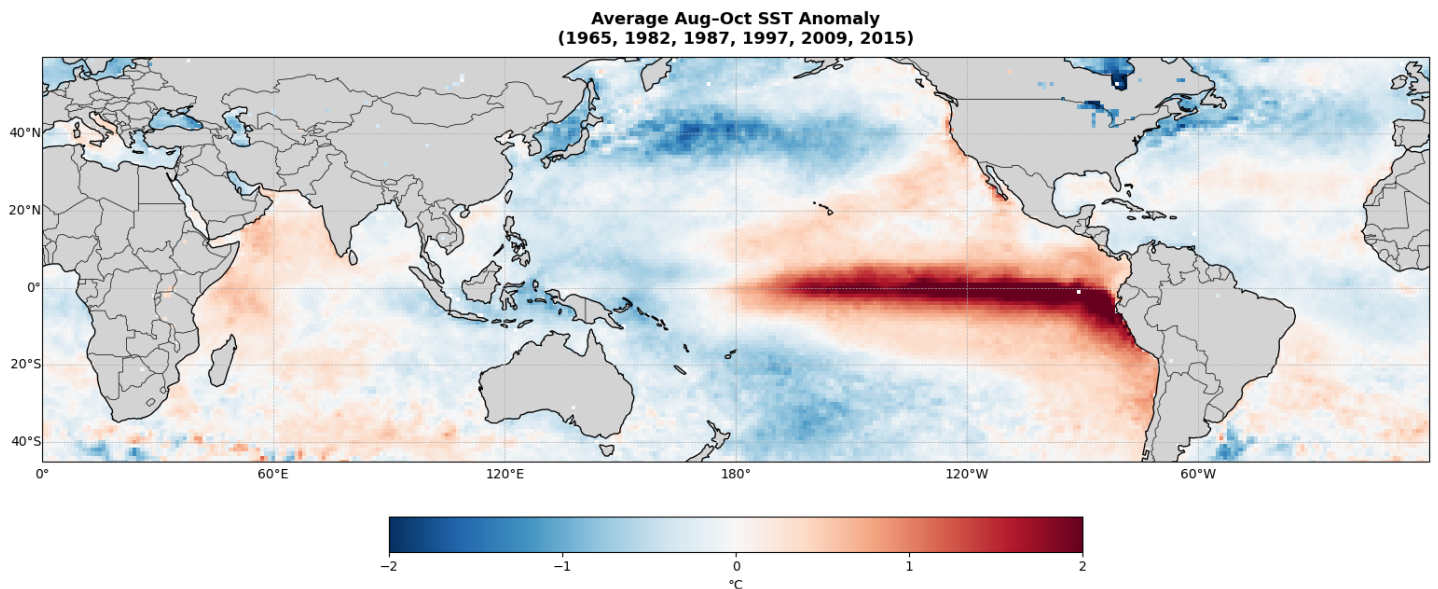


Figure 14: Average August–October SST anomalies in our six analog years.

## 2.5 ACE West of 60°W Calculation

We now explicitly forecast ACE occurring west of 60°W. While there is a relatively robust relationship between basinwide ACE and North Atlantic landfalling hurricanes (defined as hurricanes making landfall west of 60°W), there is an improved relationship between North Atlantic landfalling hurricanes and ACE west of 60°W (Figures 15 and 16) since 1979. In this analysis, we only count one landfall per storm, regardless of whether the storm made multiple landfalls at hurricane strength (e.g., Irma–2017).

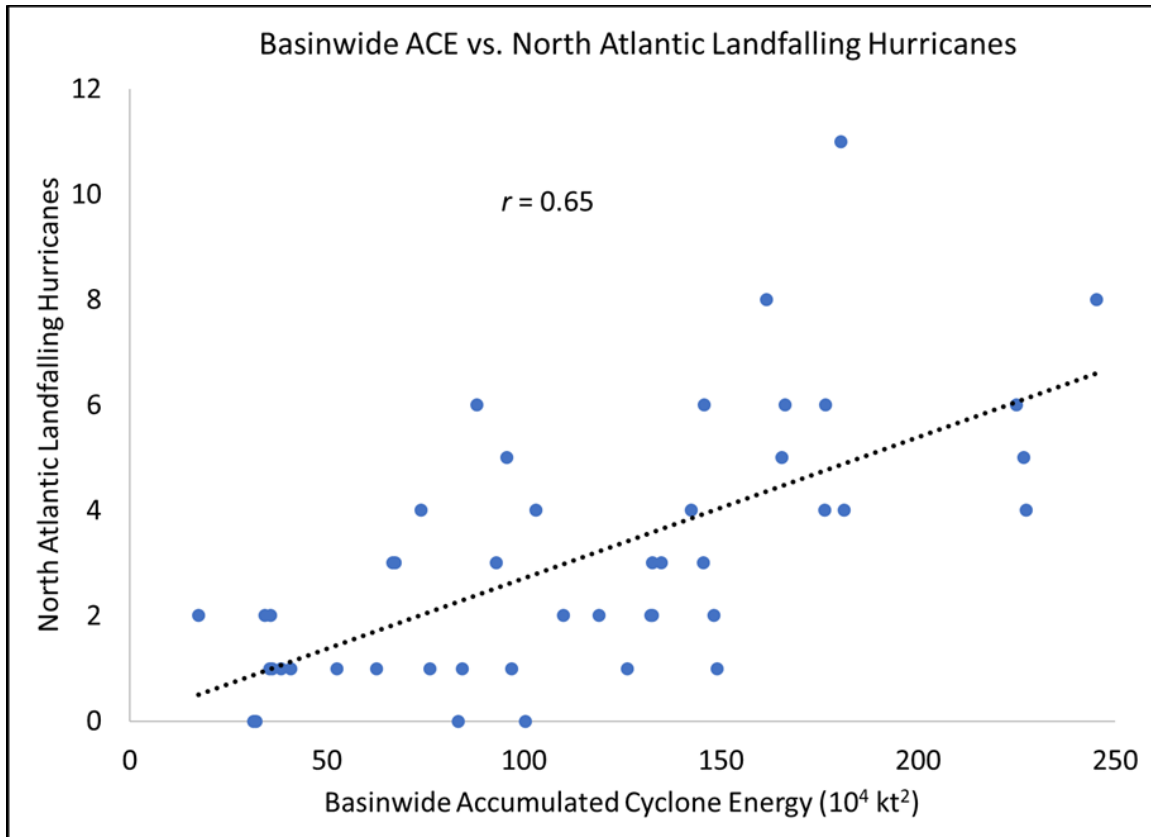


Figure 15: Scatterplot showing relationship between basinwide ACE and North Atlantic landfalling hurricanes.

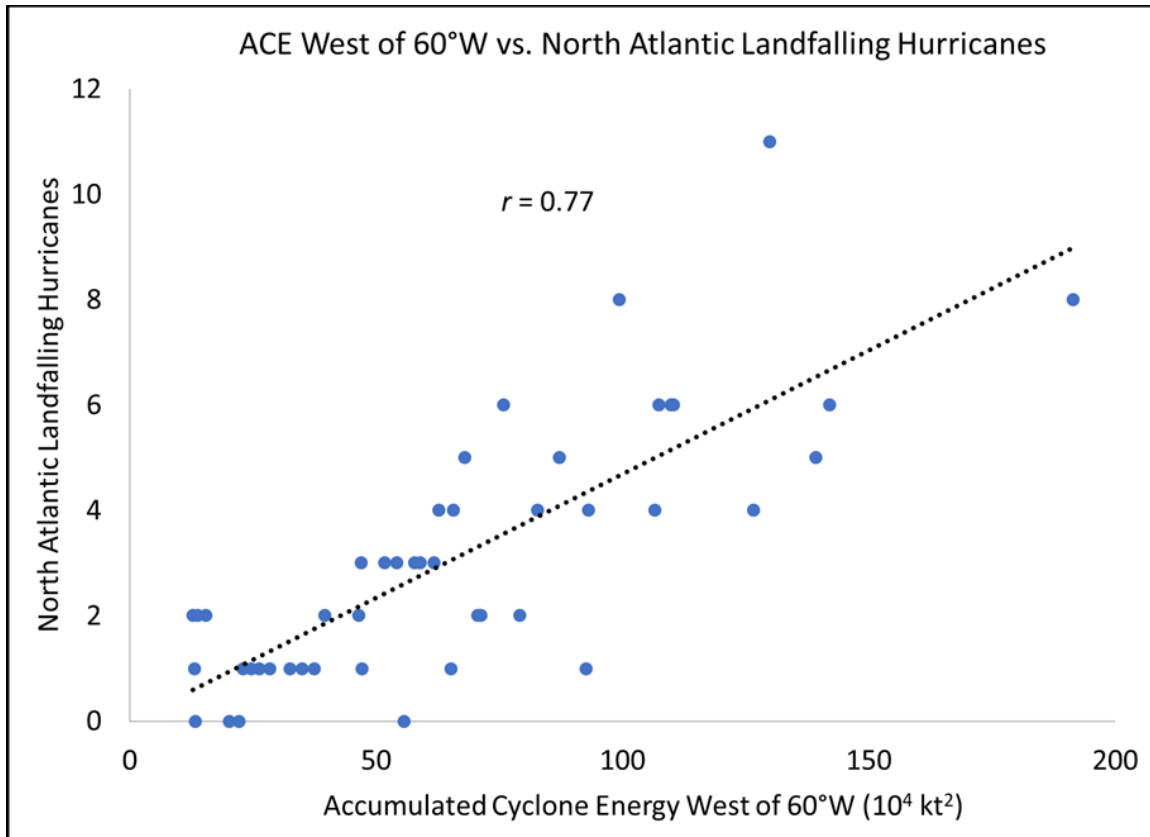


Figure 16: Scatterplot showing relationship between ACE west of 60°W and North Atlantic landfalling hurricanes.

In general, years characterized by El Niño conditions tend to have slightly less ACE west of 60°W than La Niña seasons, likely due to both more conducive conditions in the western Atlantic in La Niña seasons, as well as an increased chance of recurvature for TCs in El Niño seasons ([Colbert and Soden 2012](#); [Patricola-DiRosario et al. 2026](#)). This has certainly been the case in 2023, 2024 and 2025. In 2023, a strong El Niño occurred, the subtropical high was quite weak, and many of the TCs that occurred recurved east of 60°W. In 2024, the western Atlantic was much busier for TC formations, and 8 of the 11 hurricanes that formed made landfall west of 60°W. While there was considerable ACE generated west of 60°W in 2025, only one hurricane (Melissa), albeit one of the strongest hurricanes on record, made landfall. 48%, 61% and 74% of basinwide ACE occurred west of 60°W in 2023, 2024 and 2025, respectively.

We use data from 1979–2025 and base ENSO classifications on the August–October-averaged RONI. Years with a RONI  $\geq 0.5^\circ\text{C}$  are classified as El Niño, years with an RONI  $\leq -0.5^\circ\text{C}$  are classified as La Niña, while all other seasons are classified as neutral ENSO. The RONI index is calculated by differencing SSTs in the Nino 3.4 region (5°S–5°N, 170–120°W) from tropical mean SST (20°S–20°N, 0°–360°) and scaling the time series to match the variability of the observed SST in the Nino 3.4 region. The RONI has recently been adopted by NOAA as its official ENSO classification index.

We find that 54% of basinwide ACE occurs west of 60°W in El Niño years, while 63% of basinwide ACE occurs west of 60°W in La Niña years (Figure 17). In neutral ENSO years, 56% of basinwide ACE occurs west of 60°W. Given that we are expecting strong El Niño conditions with this outlook, we are estimating 50% of basinwide ACE to occur west of 60°W in 2026.

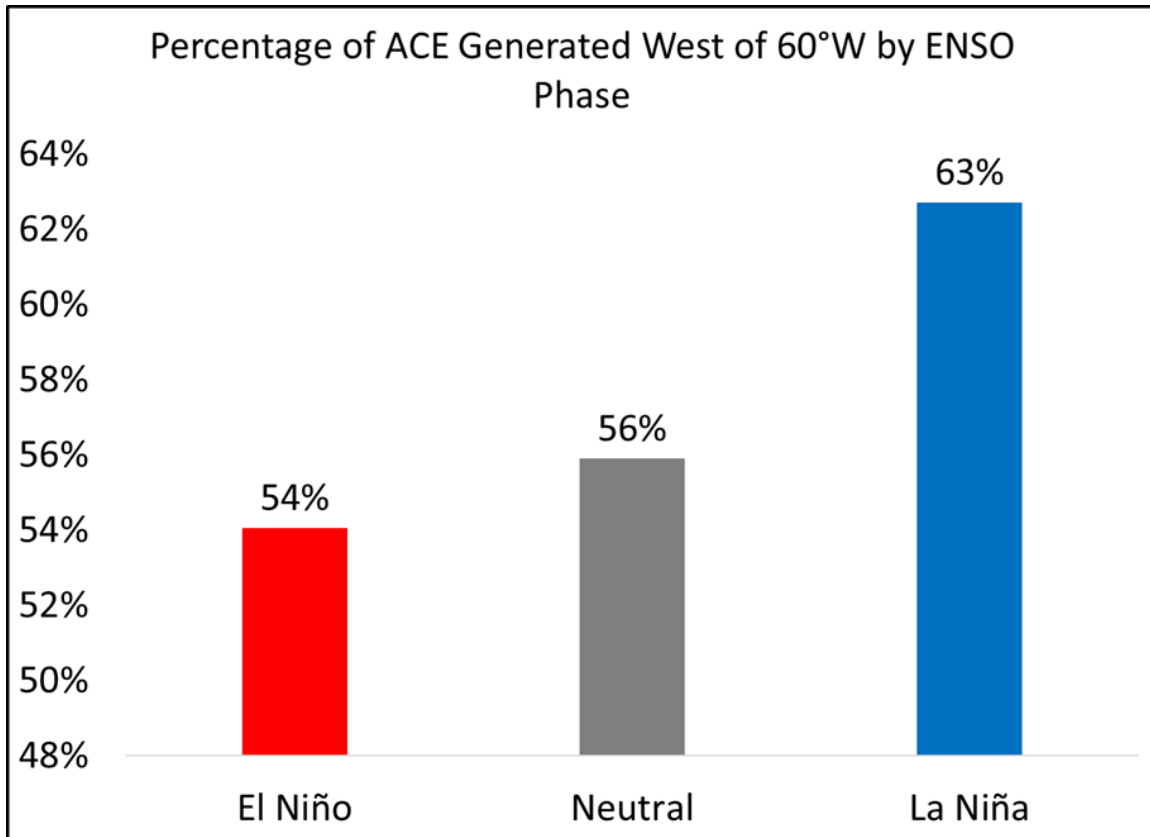


Figure 17: Percentage of ACE generated west of 60°W by ENSO phase.

## 2.5 August Forecast Summary and Final Adjusted Forecast

Table 8 shows our final adjusted early August forecast for the 2026 season which is a combination of our statistical schemes, statistical/dynamical schemes, and analog scheme as well as qualitative adjustments for other factors not explicitly contained in any of these schemes. Our final forecast is close to the average of all of these schemes.

Table 8: Summary of our early August statistical forecasts, our statistical/dynamical forecasts, our analog forecast, the average of these seven schemes and our adjusted final forecast for the 2026 hurricane season.

| Forecast Parameter and 1991–2020 Average<br>(in parentheses) | Statistical<br>Scheme | 50-Day<br>Scheme | ECMWF<br>Scheme | Met Office<br>Scheme | JMA<br>Scheme | CMCC<br>Scheme | Analog<br>Scheme | 7-Scheme<br>Average | Adjusted Final<br>Forecast |
|--------------------------------------------------------------|-----------------------|------------------|-----------------|----------------------|---------------|----------------|------------------|---------------------|----------------------------|
| Named Storms (14.4)                                          | 12.7                  | 13.0             | 8.8             | 10.0                 | 9.4           | 9.9            | 8.5              | 10.3                | 9                          |
| Named Storm Days (69.4)                                      | 47.4                  | 49.8             | 31.0            | 38.9                 | 34.7          | 38.1           | 34.6             | 39.2                | 35                         |
| Hurricanes (7.2)                                             | 4.5                   | 4.7              | 3.1             | 4.0                  | 3.5           | 3.9            | 3.2              | 3.8                 | 4                          |
| Hurricane Days (27.0)                                        | 12.9                  | 14.3             | 6.1             | 10.4                 | 8.1           | 9.9            | 10.6             | 10.3                | 12                         |
| Major Hurricanes (3.2)                                       | 1.6                   | 1.7              | 0.8             | 1.3                  | 1.0           | 1.2            | 1.3              | 1.3                 | 1                          |
| Major Hurricane Days (7.4)                                   | 2.3                   | 2.8              | 0.5             | 1.4                  | 1.0           | 1.2            | 3.0              | 1.7                 | 2                          |
| ACE Index (123)                                              | 64                    | 70               | 31              | 50                   | 40            | 48             | 52               | 51                  | 50                         |
| NTC Activity (135%)                                          | 79                    | 85               | 44              | 63                   | 53            | 61             | 62               | 64                  | 60                         |

### 3 Forecast Uncertainty

This season we continue to use probability of exceedance curves as discussed in [Saunders et al. \(2020\)](#) to quantify forecast uncertainty. In that paper, we outlined an approach that uses statistical modeling and historical skill of various forecast models to arrive at a probability that particular values of hurricane numbers and ACE would be exceeded. Here we display probability of exceedance curves for hurricanes and ACE (Figures 18 and 19), using the error distributions calculated from both normalized cross-validated statistical as well as the cross-validated statistical/dynamical hindcasts from SEAS5. Hurricane numbers are fit to a Poisson distribution, while ACE is fit to a Weibull distribution. Table 9 displays one standard deviation uncertainty ranges (~68% of all forecasts within this range). This uncertainty estimate is also very similar to the 70% uncertainty range that NOAA provides with its forecasts. We use Poisson distributions for all storm parameters (e.g., named storms, hurricanes and major hurricanes), while we use a Weibull distribution for all integrated parameters except for major hurricane days (e.g., named storm days, ACE, etc.). We use a Laplace distribution for major hurricane days.

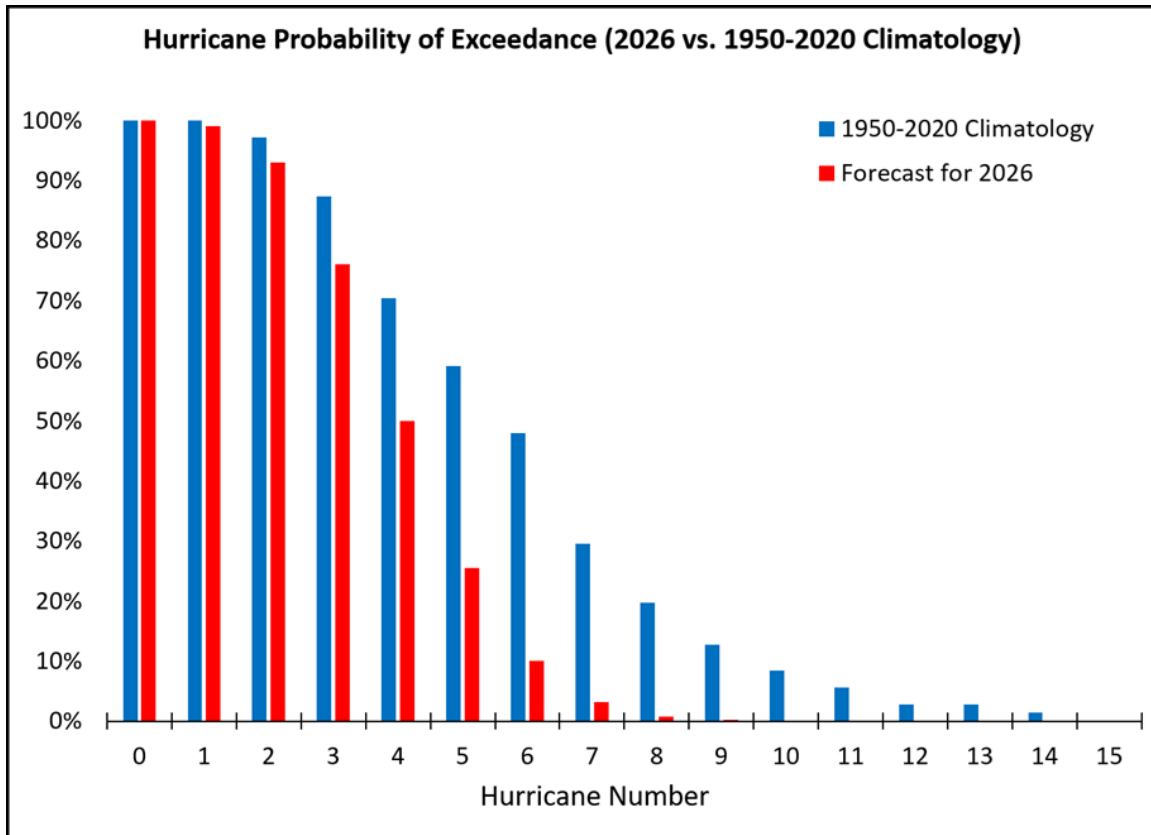


Figure 18: Probability of exceedance plot for hurricane numbers for the 2026 Atlantic hurricane season. The values on the x-axis indicate that the number of hurricanes exceeds that specific number. For example, 97% of Atlantic hurricane seasons from 1950–2020 have had more than two hurricanes.

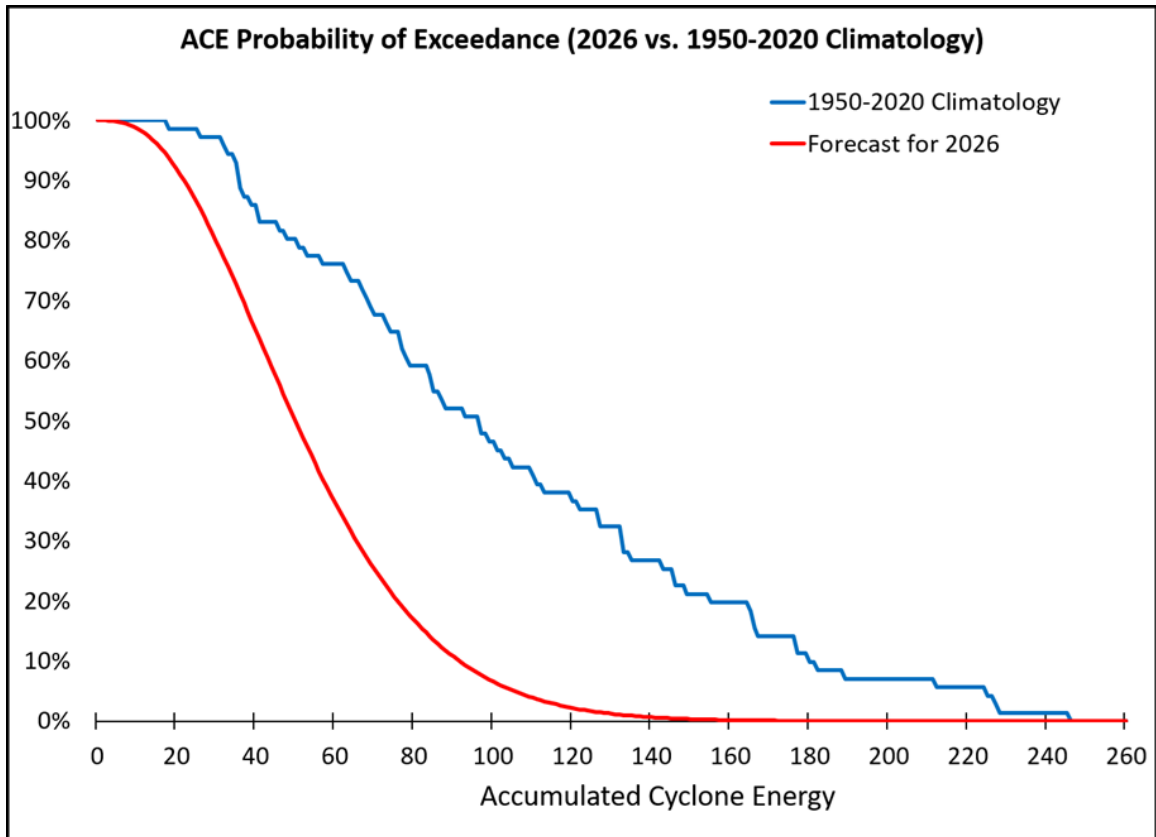


Figure 19: As in Figure 18 but for ACE.

Table 9: Forecast ranges for each parameter. Note that the forecast spread may not be symmetric around the mean value, given the historical distribution of tropical cyclone activity.

| Parameter                           | 2026 Forecast | Uncertainty Range (68% of Forecasts Likely to Fall in This Range) |
|-------------------------------------|---------------|-------------------------------------------------------------------|
| Named Storms (NS)                   | 9             | 7 – 11                                                            |
| Named Storm Days (NSD)              | 35            | 21 – 51                                                           |
| Hurricanes (H)                      | 4             | 2 – 6                                                             |
| Hurricane Days (HD)                 | 12            | 6 – 20                                                            |
| Major Hurricanes (MH)               | 1             | 0 – 2                                                             |
| Major Hurricane Days (MHD)          | 2             | 0 – 5                                                             |
| Accumulated Cyclone Energy (ACE)    | 50            | 26 – 83                                                           |
| ACE West of 60°W                    | 25            | 12 – 45                                                           |
| Net Tropical Cyclone (NTC) Activity | 60            | 34 – 94                                                           |

## 4 ENSO

Over the past several weeks, El Niño has continued to intensify, with recent values of the Relative Niño 3.4 index approaching the strong El Niño range (1.5–1.9°C) (Figure 20). Figure 21 displays the locations of the various Niño regions displayed in Figure 20. Please note that all SST anomalies in the various Niño regions now are

calculated as relative SST anomalies, that is, all SST anomalies are calculated relative to the tropical average SST (20°S–20°N, 0–360°). These relative SST anomalies have been adopted as part of NOAA’s shift in February 2026 to the RONI.

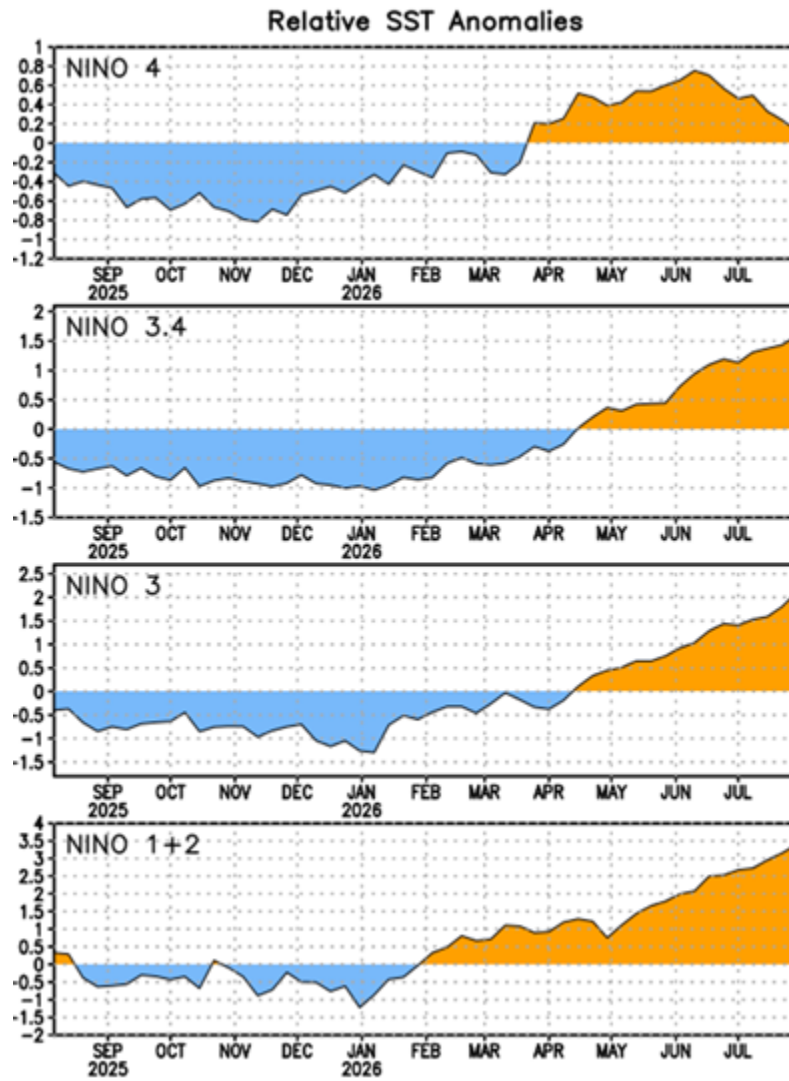


Figure 20: SST anomalies for several ENSO regions over the past year. Figure courtesy of the Climate Prediction Center.

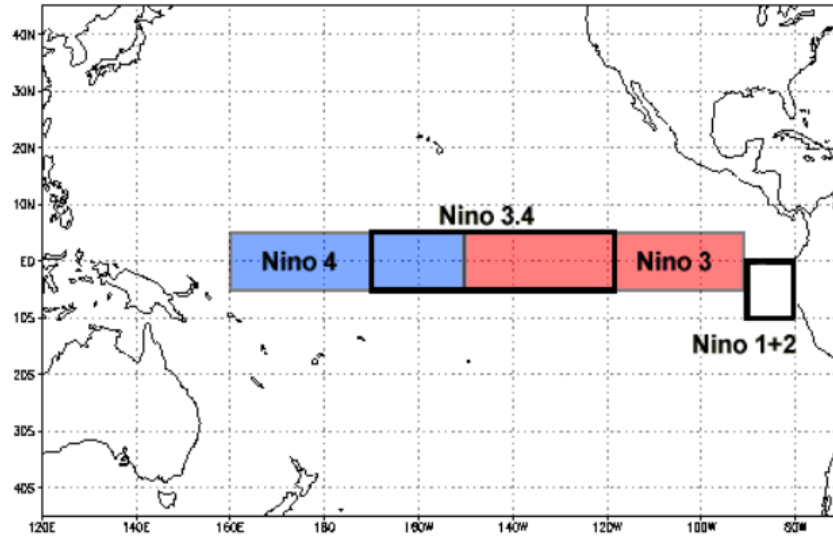


Figure 21: Location of ENSO SST regions used in Figure 19. Figure courtesy of the National Centers for Environmental Information.

Equatorial upper-ocean heat content anomalies in the eastern and central tropical Pacific have continued to increase over the past few weeks (Figure 22). Current SST anomalies are well above their long-term averages across the entire equatorial eastern and central Pacific (Figure 23).

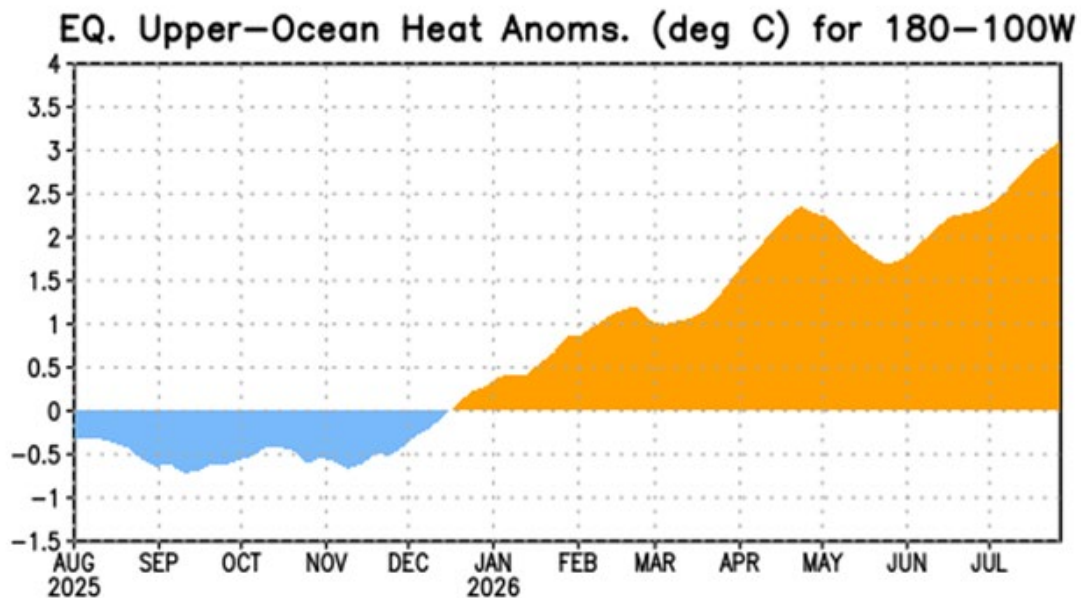


Figure 22: Central and eastern equatorial Pacific upper ocean (0–300 meters) heat content anomalies over the past year. Figure courtesy of Climate Prediction Center.

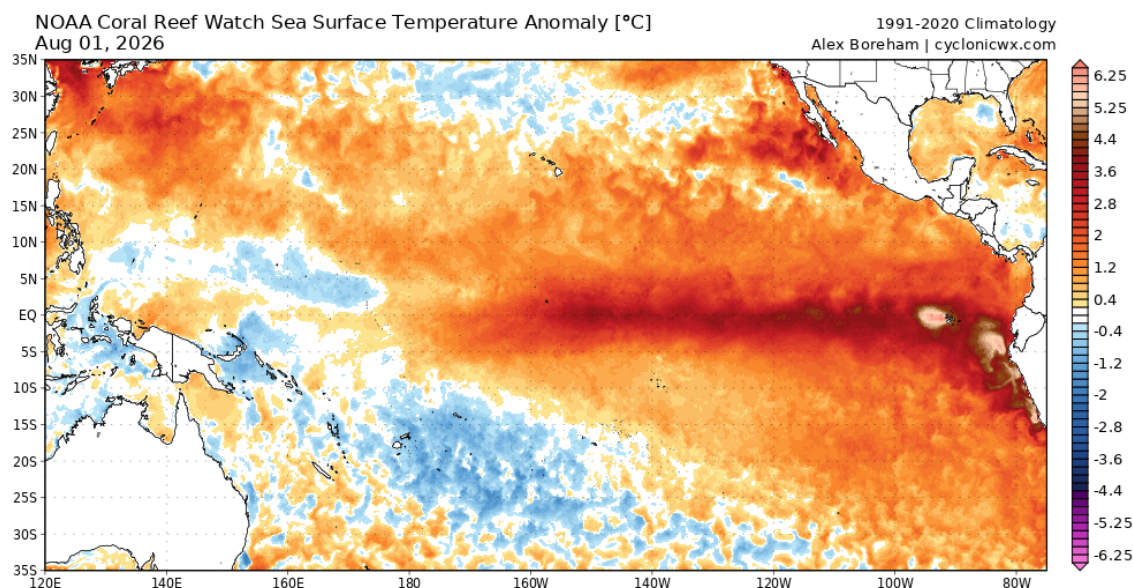


Figure 23: Current SST anomalies across the tropical and subtropical Pacific.

Table 10 displays June and July relative SST anomalies for several Nino regions. As noted earlier, over the past month, relative SST anomalies have generally increased across the eastern and central tropical Pacific, with some anomalous cooling near the International Date Line.

Table 10: June and July relative SST anomalies for Nino 1+2, Nino 3, Nino 3.4, and Nino 4, respectively. July minus June relative SST anomaly differences are also provided.

| Region   | June Relative SST<br>Anomaly (°C) | July Relative SST<br>Anomaly (°C) | July – June<br>Relative SST Anomaly (°C) |
|----------|-----------------------------------|-----------------------------------|------------------------------------------|
| Nino 1+2 | +2.4                              | +2.9                              | +0.5                                     |
| Nino 3   | +1.2                              | +1.8                              | +0.6                                     |
| Nino 3.4 | +1.1                              | +1.4                              | +0.3                                     |
| Nino 4   | +0.8                              | +0.4                              | -0.4                                     |

A robust downwelling (warming) oceanic Kelvin wave has caused considerable anomalous warming across the eastern tropical Pacific over the past few weeks (Figure 24). Continued anomalous warming is also expected in the next few weeks due to the lack of trade winds across the central and eastern tropical Pacific (e.g., reduced evaporational cooling), with pronounced low-level westerly wind anomalies forecast to predominate across the tropical Pacific (Figure 25).

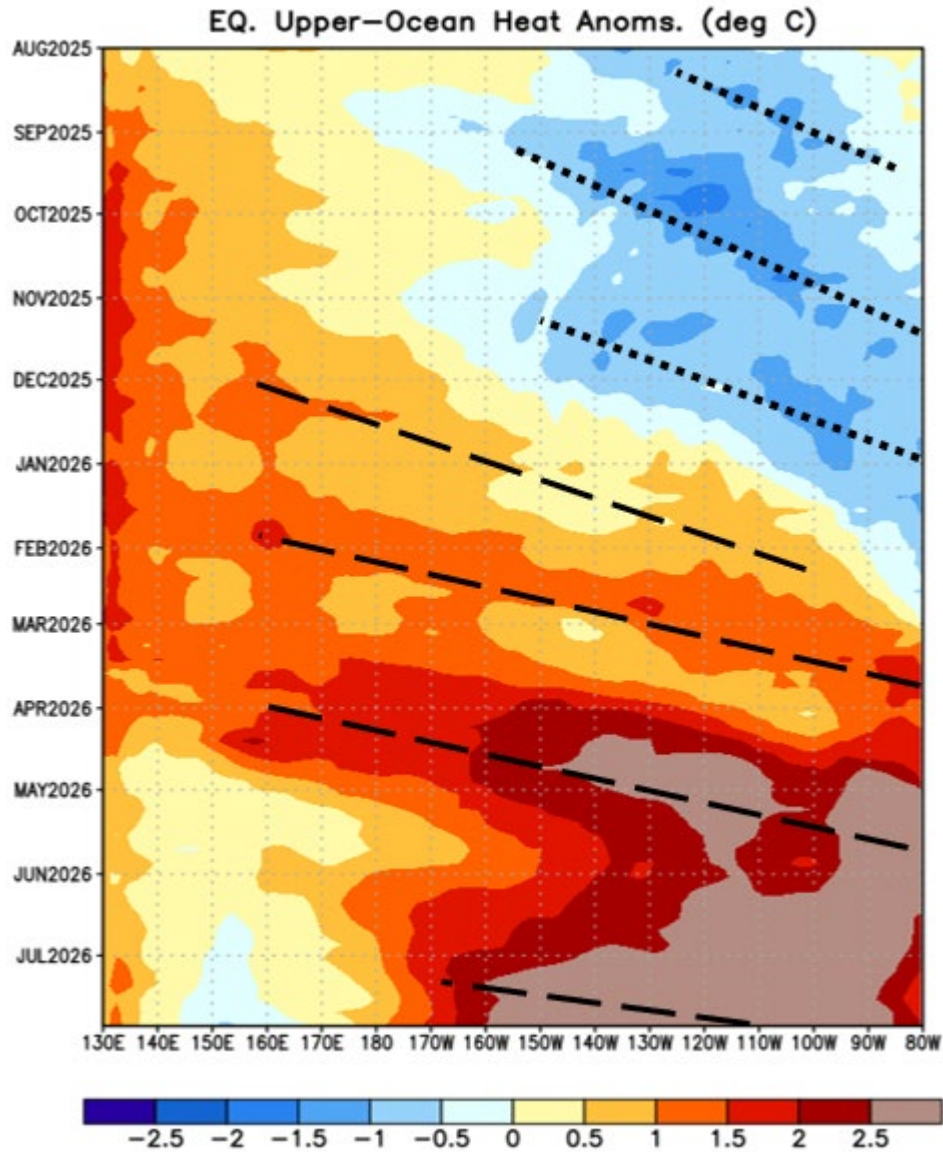


Figure 24: Upper-ocean (0–300 meter) heat content anomalies in the tropical Pacific since July 2025. Long dashed lines indicate downwelling Kelvin waves, while short dashed lines indicate upwelling Kelvin waves. Downwelling Kelvin waves result in upper-ocean heat content increases, while upwelling Kelvin waves result in upper-ocean heat content decreases. Figure courtesy of NOAA/Climate Prediction Center.

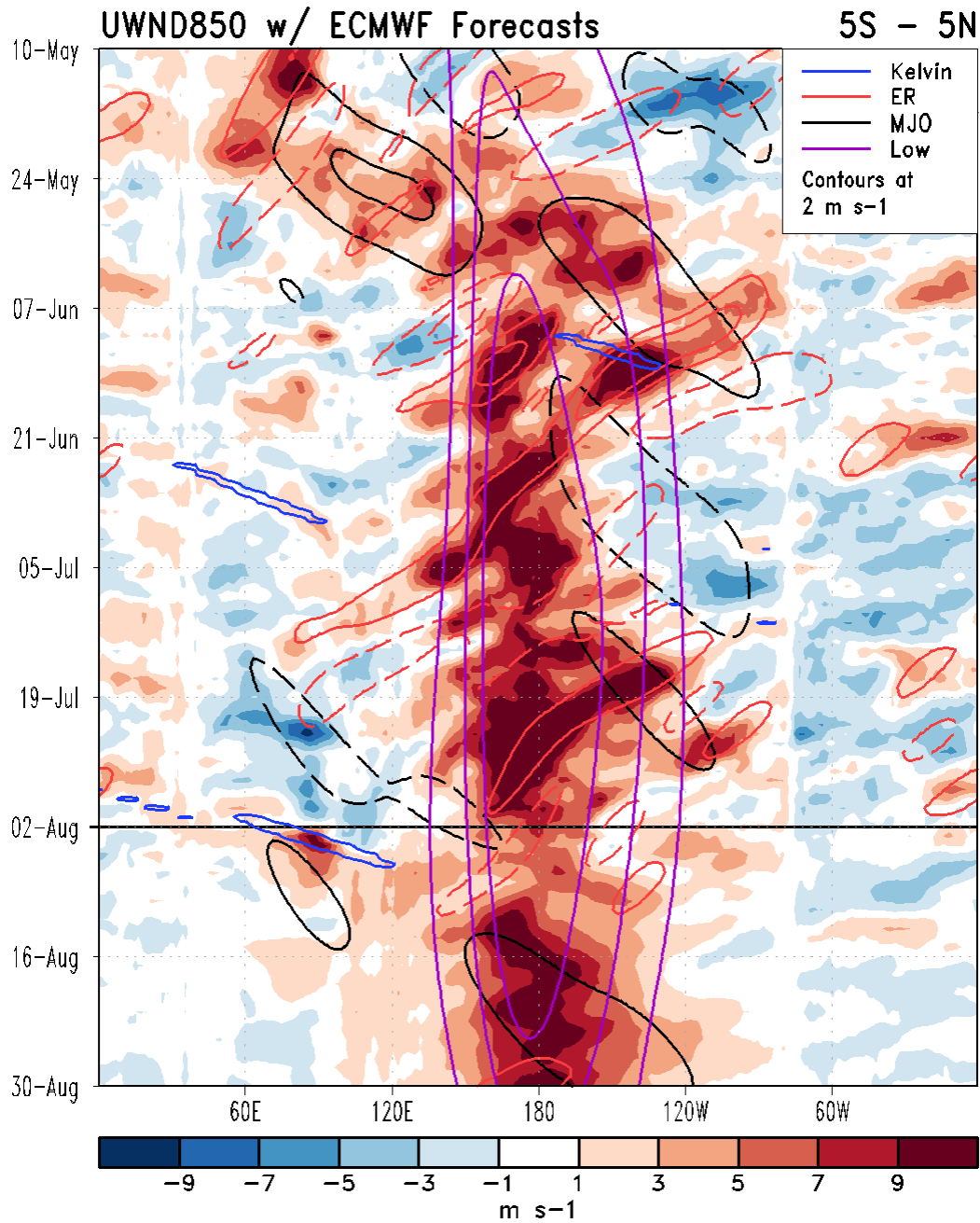


Figure 25: ECMWF observed and forecast 850-hPa zonal equatorial winds for the next 46 days. Figure courtesy of Nick Novella (NOAA/Climate Prediction Center).

While there is still some uncertainty as to how strong El Niño will be for the peak of the 2026 Atlantic hurricane season, the latest ensemble plume from ECMWF calls for an extremely high likelihood of a very strong ( $\geq 2.0^{\circ}\text{C}$ ) event, with all ensemble members exceeding  $2^{\circ}\text{C}$  by September (Figure 26).

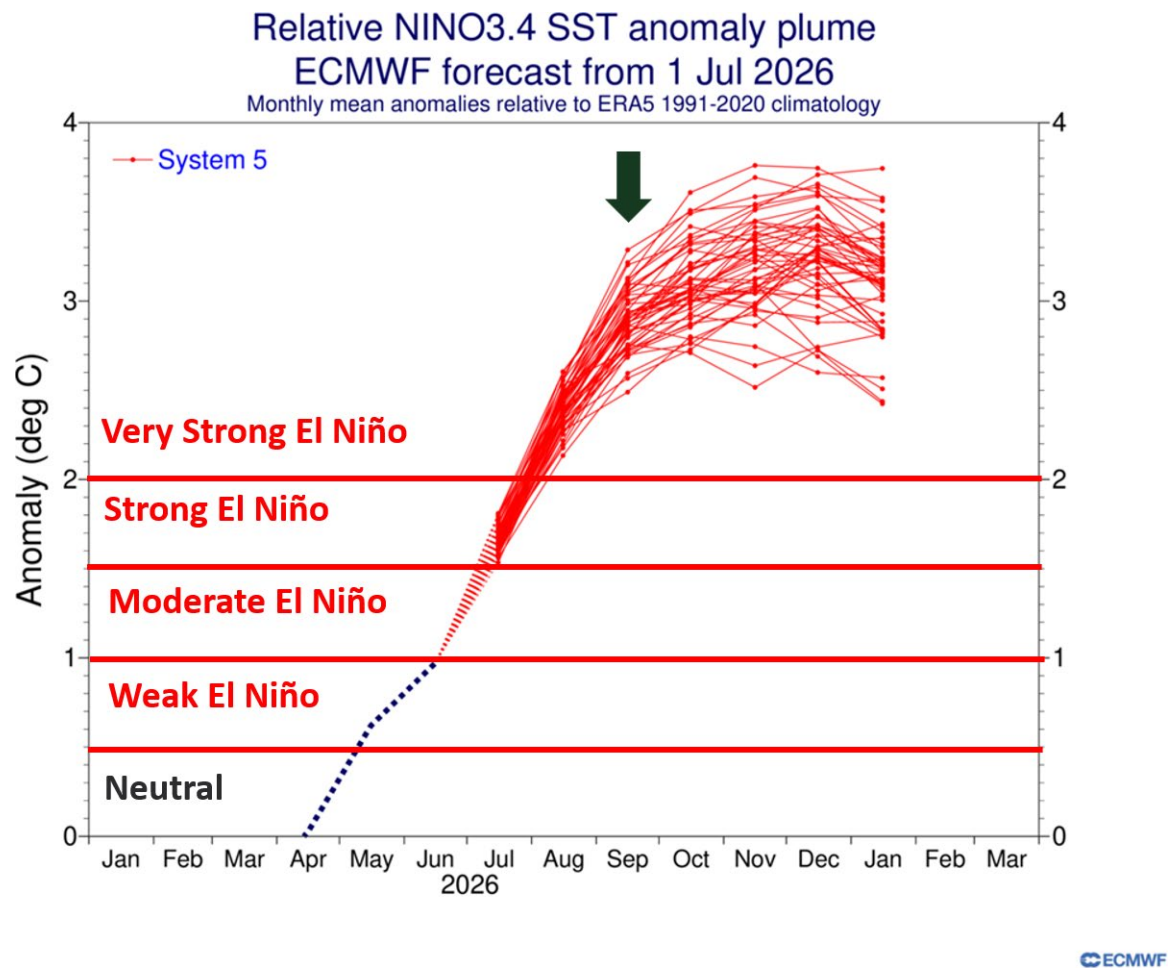


Figure 26: ENSO forecasts from the North American Multi-Model Ensemble for the RONI based on a 1 June initialization. The black arrow delineates the peak of the Atlantic hurricane season (August–October).

The latest official forecast from NOAA favors a strong El Niño event for August–October, with a 90% chance that the RONI will be  $\geq 1.5^{\circ}\text{C}$  during the three-month period (Figure 27).

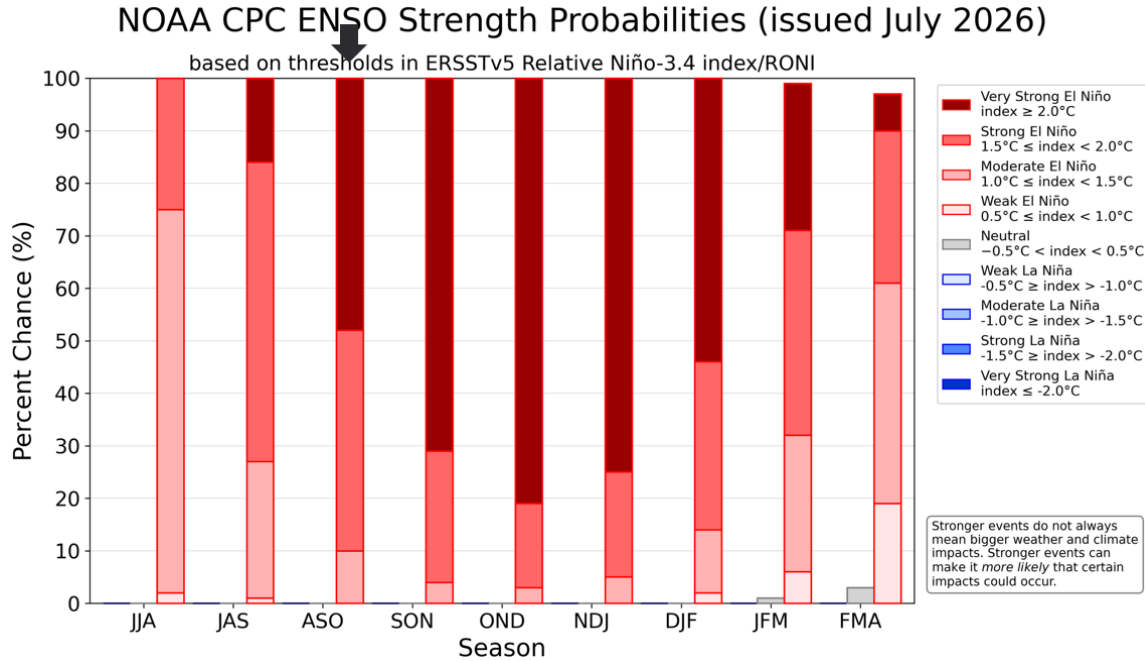


Figure 27: Official probabilistic ENSO forecast from NOAA. The black arrow delineates the peak of the Atlantic hurricane season (August–October).

Based on the above information, our best estimate is that we will have a strong ( $1.5\text{--}2.0^{\circ}\text{C}$ ) or very strong ( $\geq 2.0^{\circ}\text{C}$ ) El Niño for the peak of the Atlantic hurricane season. We believe that the odds of a strong El Niño have continued to increase over the past few weeks, likely significantly increasing tropical Atlantic and Caribbean vertical wind shear and reducing Atlantic hurricane activity.

## 5 Current Atlantic Basin Conditions

Currently, SSTs are slightly warmer than normal across the Atlantic's Main Development Region ( $10\text{--}20^{\circ}\text{N}$ ,  $85\text{--}20^{\circ}\text{W}$ ) (Figure 28). Over the past several weeks, SSTs across the Main Development Region have anomalously warmed, likely due to increased subsidence, as evidenced by anomalously high mean sea level pressure (Figure 29) associated with the El Niño. Subsidence and dryness allows for increased solar radiation reaching the ocean surface, allowing for anomalous warming. 30-day-averaged SSTs across the Main Development Region are currently tracking between what is typically experienced in an above normal and a hyperactive Atlantic hurricane season (Figure 30).

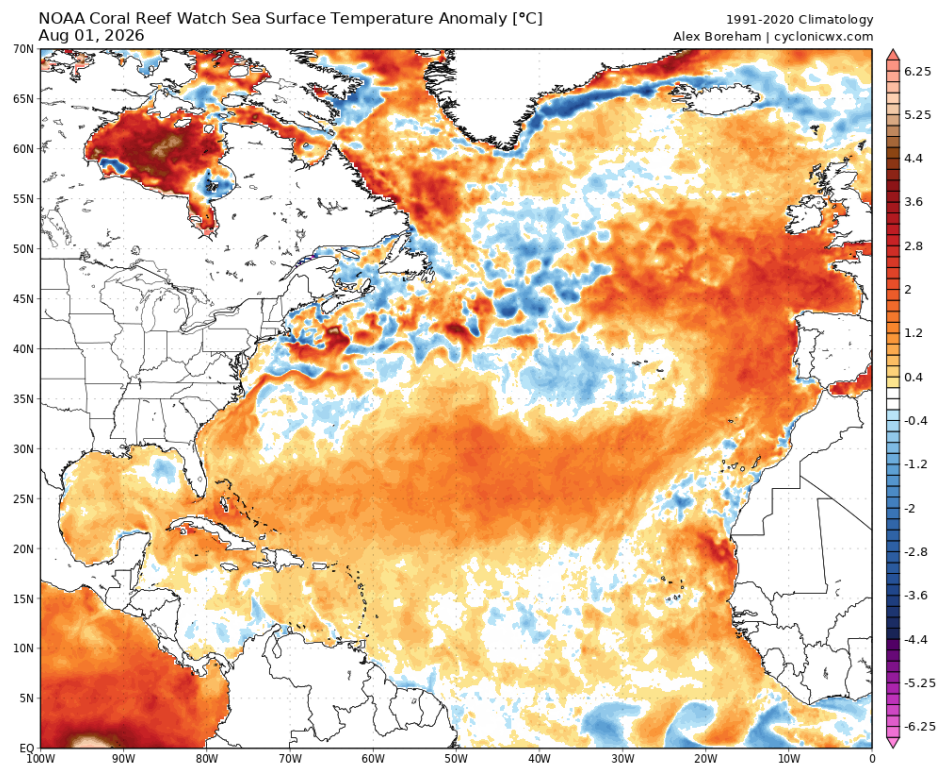


Figure 28: Current North Atlantic SST anomalies.

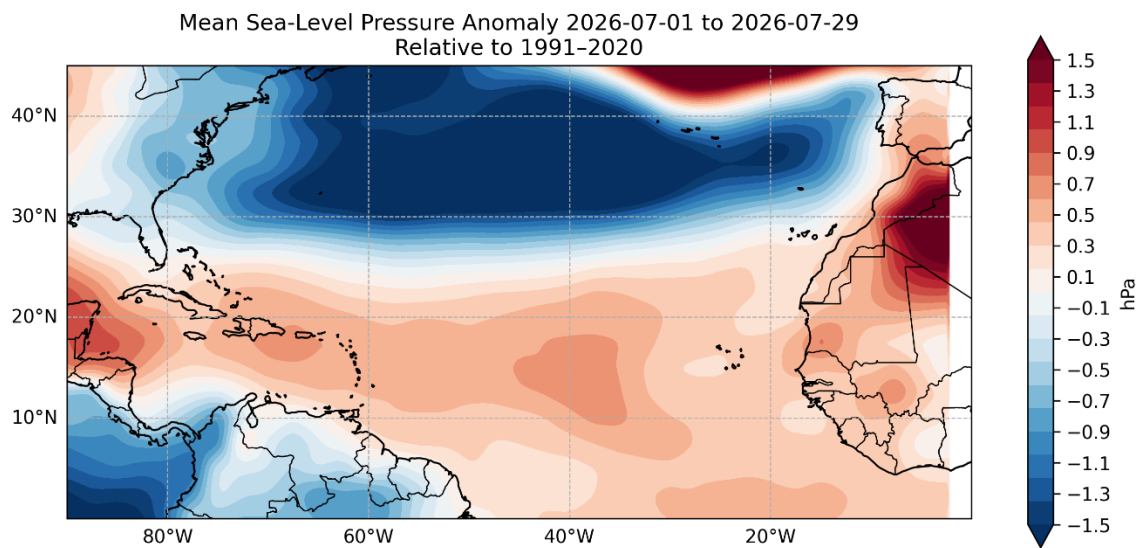


Figure 29: Anomalous North Atlantic mean sea level pressure since July 1<sup>st</sup>.

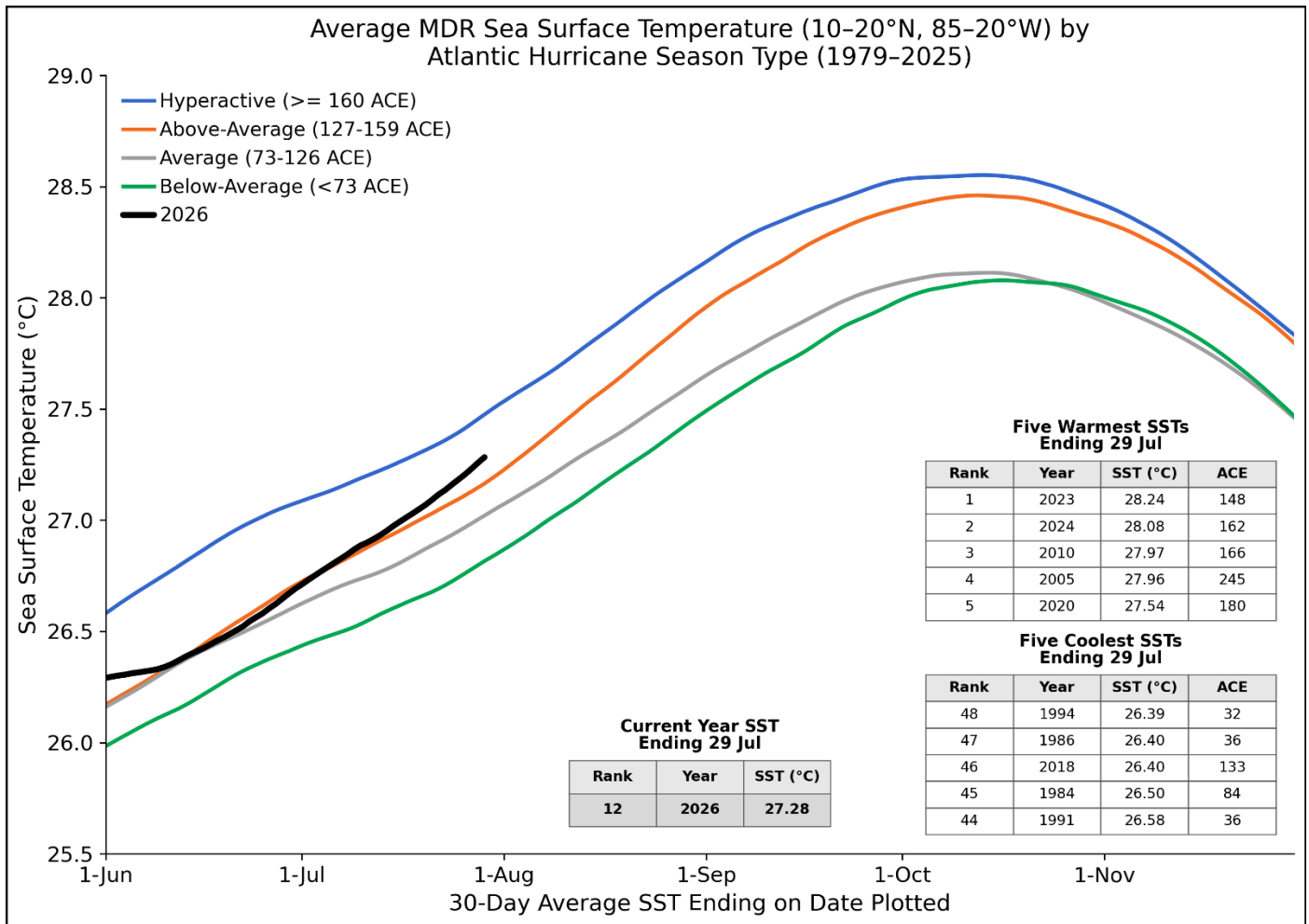


Figure 30: 30-day average SSTs averaged across the Main Development Region for various Atlantic hurricane season types from 1979–2025 based on NOAA’s definition. Sea surface temperature anomalies in the tropical Atlantic in 2026 are tracking between an above-average and hyperactive Atlantic hurricane season.

Figure 31 shows the forecast for the next ~4 weeks of low-level winds across the Atlantic. Trade winds are generally forecast to be stronger than normal for the next few weeks, which we anticipate will likely keep the Atlantic Main Development Region from too much additional anomalous warming by the peak of the Atlantic hurricane season. While the current SST anomaly pattern correlates relatively well with what is typically seen in August of active Atlantic hurricane seasons (Figure 32), we anticipate that the strong to very strong El Niño will dominate the tropical circulation.

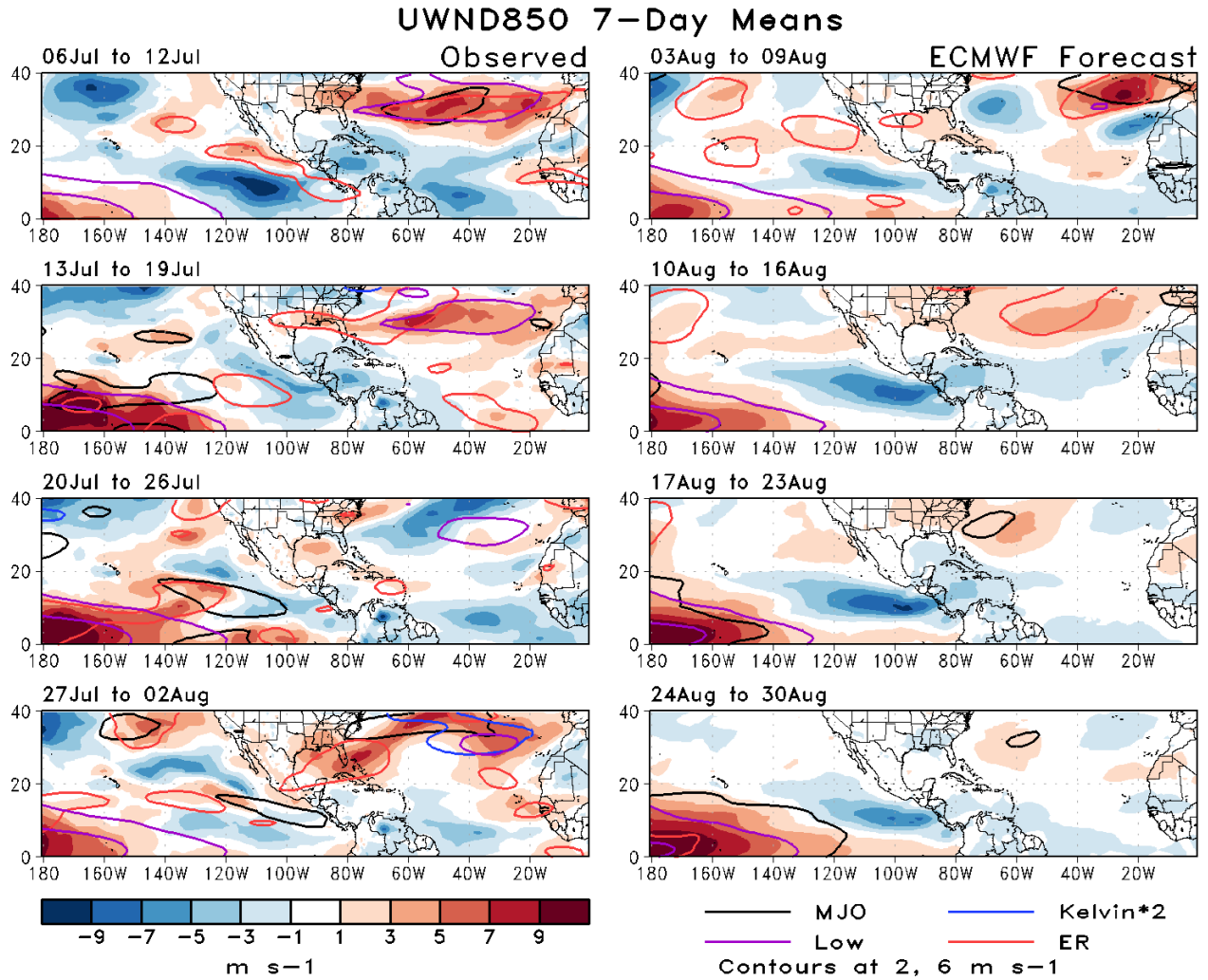


Figure 31: Observed low-level zonal wind anomalies across portions of the Western Hemisphere over the past four weeks and predicted low-level zonal winds by ECMWF through 30 August. Figure courtesy of Nick Novella (NOAA/Climate Prediction Center).

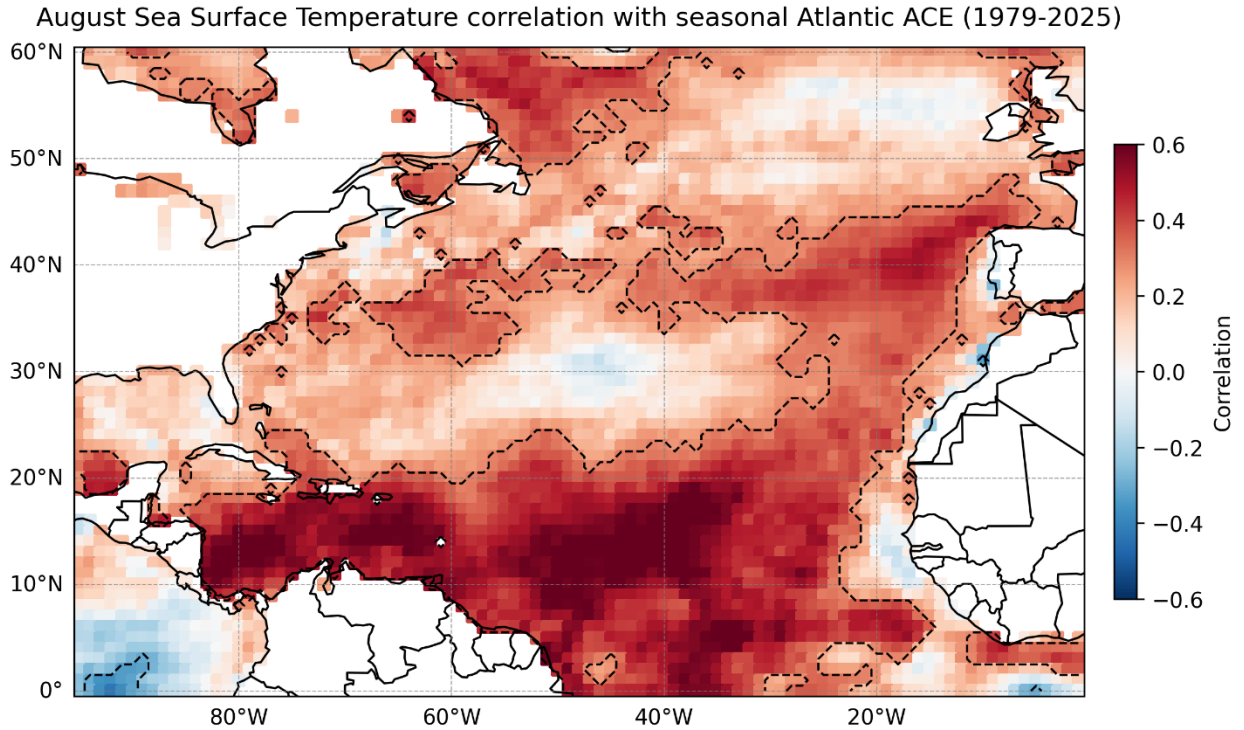


Figure 32: Correlations between August sea surface temperatures in the North Atlantic and annual Atlantic ACE from 1979–2025.

The tropical Atlantic has had extremely high levels of vertical wind shear over the past few weeks, with vertical wind shear over the past 30 days well above normal across the entire Main Development Region (Figure 33). Western tropical Atlantic (10–20°N, 85–50°W) vertical wind shear correlates quite strongly with Atlantic ACE, with 30-day-average correlations growing strongly between late June and mid-July (Figure 34). Currently, western tropical Atlantic vertical wind shear is at its second highest on record, trailing only 2015 (Figure 35).

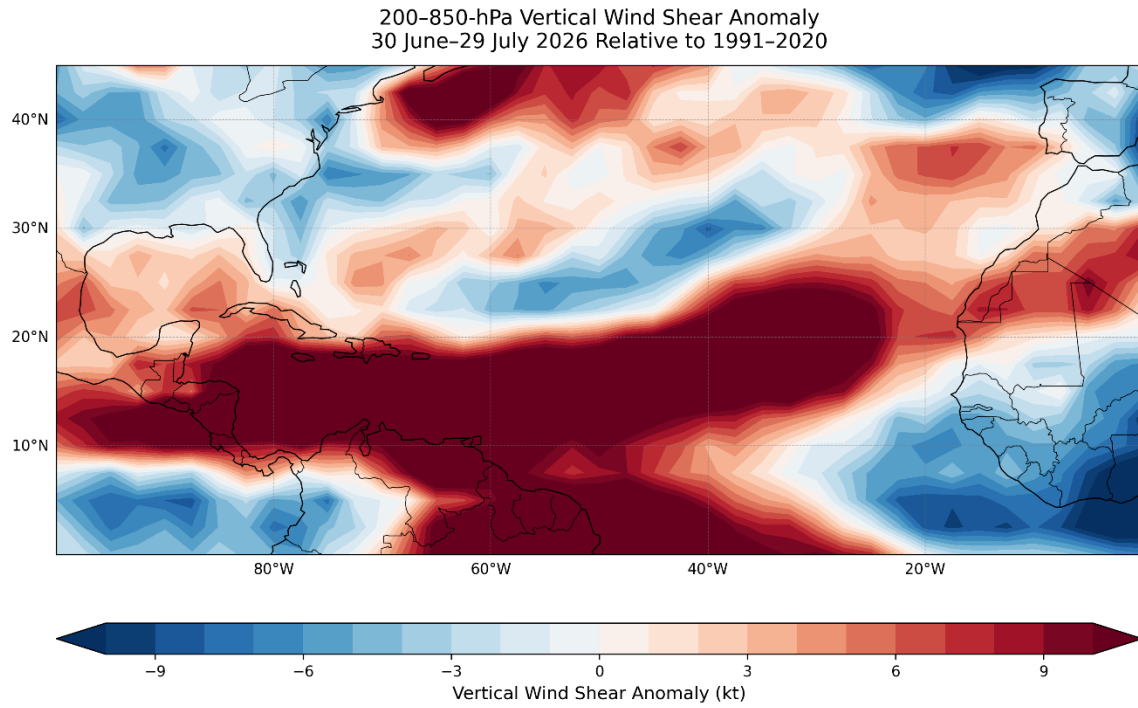


Figure 33: 30 June – 29 July-averaged zonal vertical wind shear anomalies across the tropical and subtropical Atlantic.

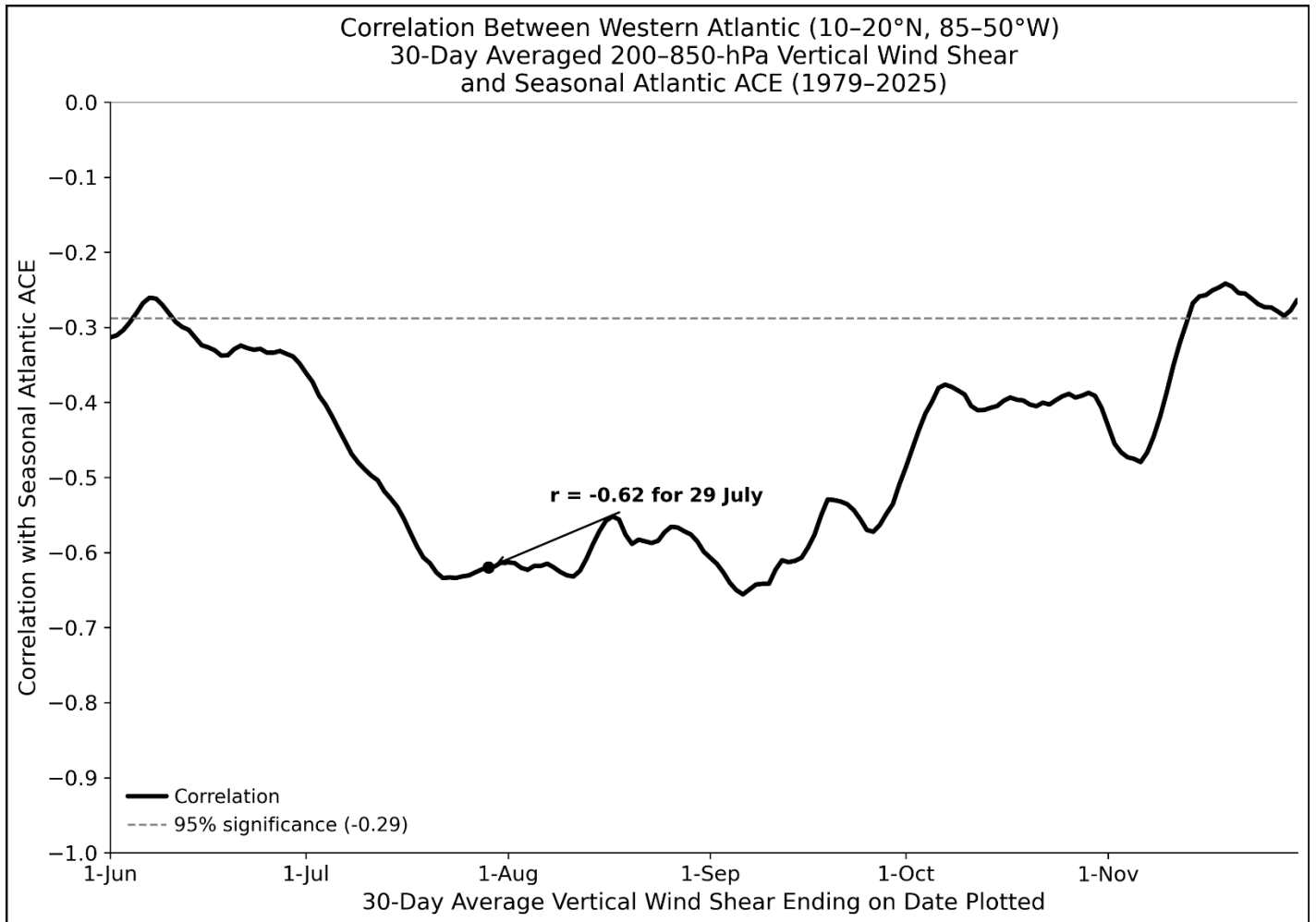


Figure 34: 30-day average correlations between western tropical Atlantic zonal vertical wind shear and seasonal Atlantic ACE. The correlation with the final day of the most recent 30-day average is displayed on the figure.

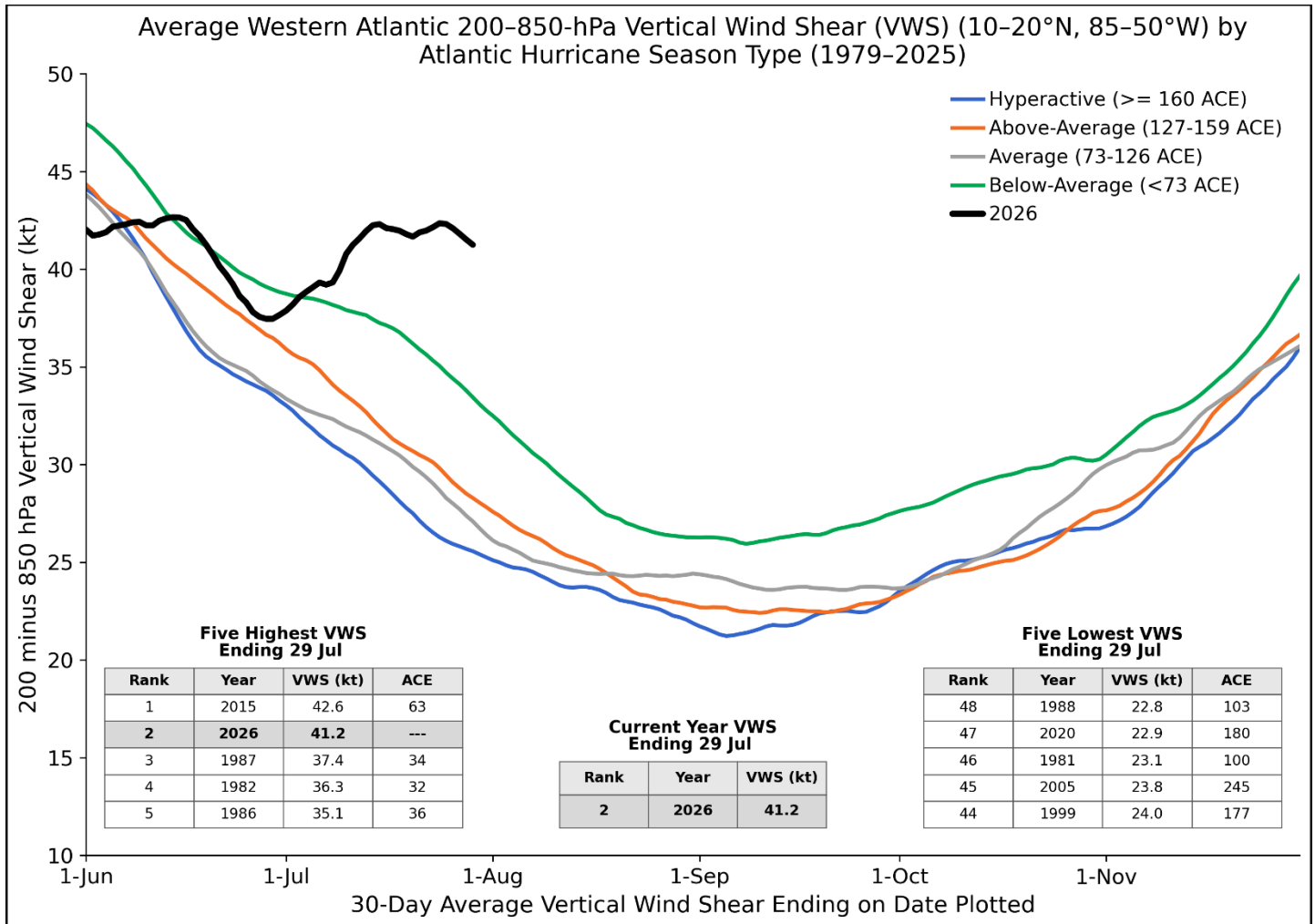


Figure 35: 30-day average western tropical Atlantic zonal vertical wind shear for various Atlantic hurricane season types from 1979–2025 based on NOAA’s definition. Western tropical Atlantic zonal vertical wind shear anomalies are currently the 2<sup>nd</sup> highest on record, trailing only 2015.

ECMWF is forecasting continued high levels of vertical wind shear across the Main Development Region, with especially high anomalies across the western tropical Atlantic over the next several weeks (Figure 36). These high vertical wind shear anomalies are a classic signal of a strong El Niño event.

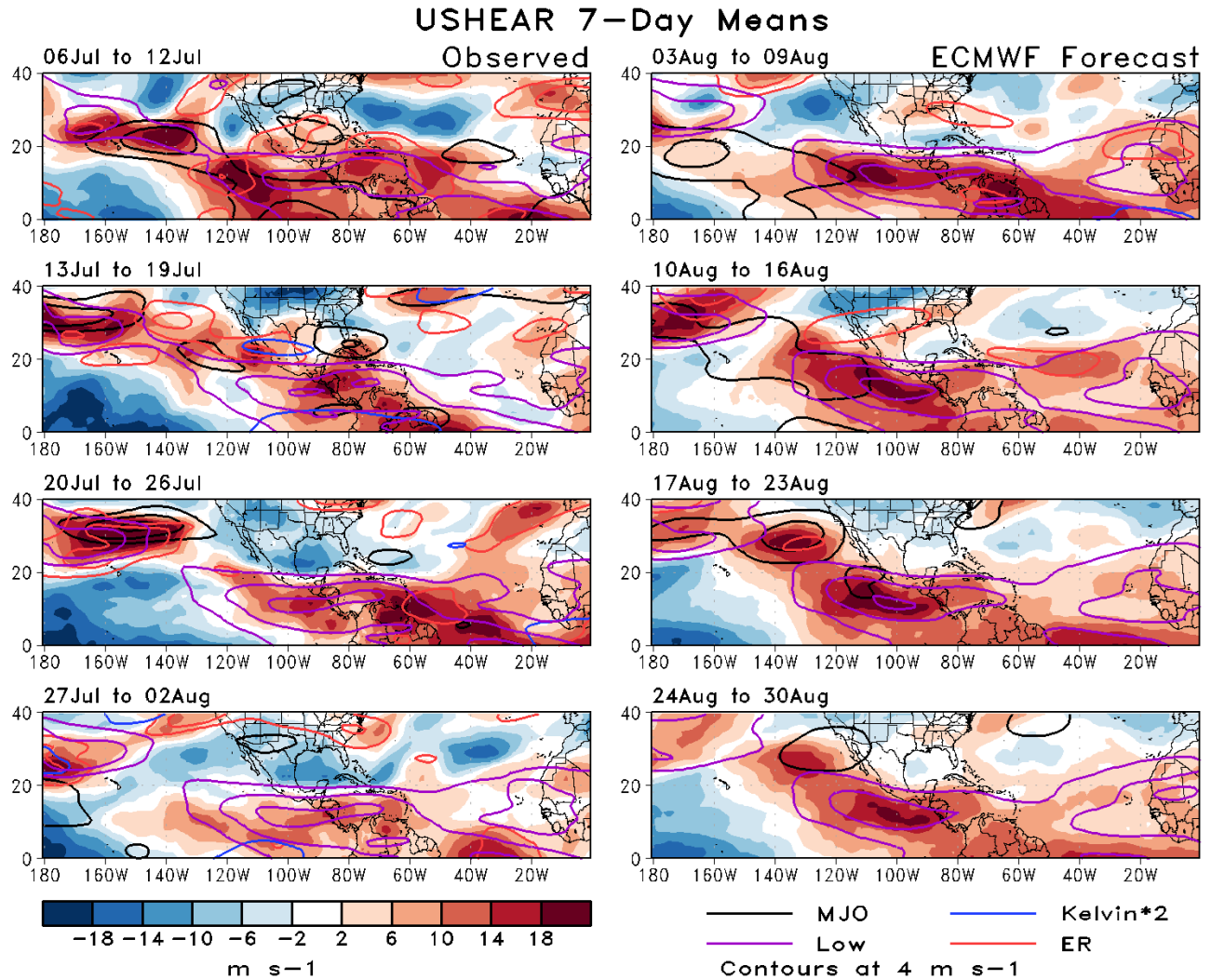


Figure 36: Observed 200 minus 850 hPa zonal wind anomalies and predicted 200 hPa minus 850 hPa zonal wind anomalies by ECMWF through 30 August. Figure courtesy of Nick Novella (NOAA/Climate Prediction Center).

So far, tropical Atlantic dust has been running somewhat below normal across the tropical Atlantic (10–20°N, 60–20°W) (Figure 37). Higher tropical Atlantic dust during June/July typically is associated with quieter Atlantic hurricane seasons, while lower tropical Atlantic dust during June/July is associated with busier hurricane seasons. Despite the relatively low dust so far in 2026, we anticipate a well below-normal season given the previously discussed high levels of vertical wind shear that we have observed and anticipate to continue across the tropical Atlantic and Caribbean.

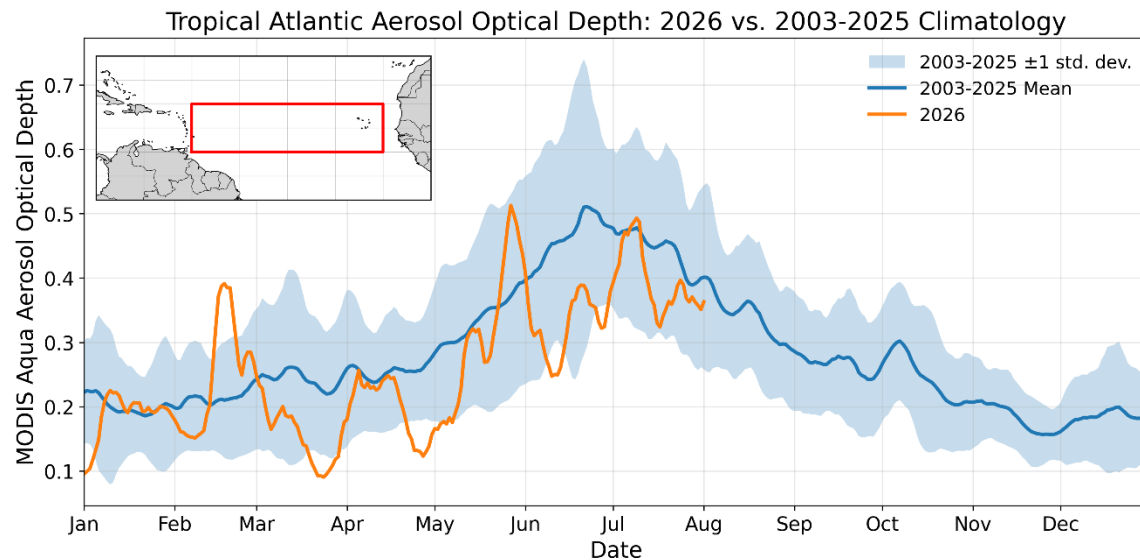


Figure 37: Tropical Atlantic aerosol optical depth at 550 nm in 2026 compared with the 2003–2025 average.

## 6 West Africa Conditions

Over the past several weeks, monsoon rainfall over West Africa has been somewhat below average (Figure 38). Below-average monsoon rainfall over West Africa frequently occurs with El Niño events. This relative lack of precipitation indicates that African easterly waves have likely had relatively weak amplitudes so far in 2026. Weaker amplitude African easterly waves are less likely to become TCs.

## ARC2 30-Day Total Rainfall Anomaly (mm)

Period: 03Jul2026 – 01Aug2026

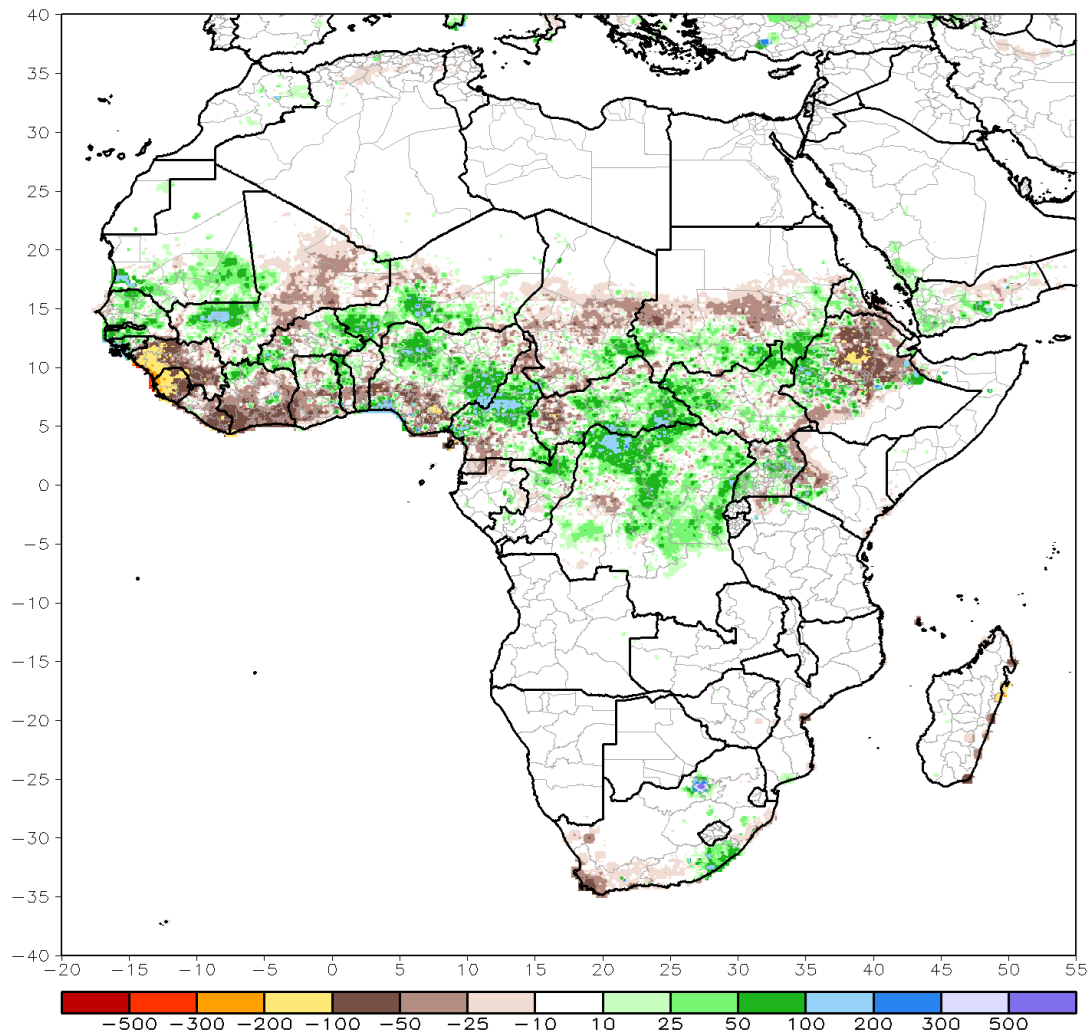


Figure 38: Observed precipitation across Africa from 3 July – 1 August, based on the African Rainfall Estimation Algorithm Version 2.

## 6 Tropical Cyclone Impact Probabilities for 2026

In addition to our basin-wide forecast, we calculate the impacts of tropical cyclones for each state and county/parish along the Gulf and East Coasts, tropical cyclone-prone provinces of Canada, states in Mexico, islands in the Caribbean and countries in Central America. We have used NOAA's Historical Hurricane Tracks [website](#) and selected all named storms, hurricanes and major hurricanes that have tracked within 50 miles of each landmass from 1880–2020. This approach allows for tropical cyclones that may have made landfall in an immediately adjacent region to be counted for all regions that were in close proximity to the landfall location of the storm. We then fit

the observed frequency of storms within 50 miles of each landmass using a Poisson distribution to calculate the climatological odds of one or more events within 50 miles.

Net landfall probability is shown to be linked to overall Atlantic basin ACE. Long-term statistics show that, on average, the more active the overall Atlantic basin hurricane season is, the greater the probability of hurricane landfalls for various landmasses in the basin. We adjust landfall probabilities based on the ratio of predicted ACE west of 60°W to the average ACE west of 60°W, as almost all landmasses that we are issuing probabilities for are west of 60°W.

Table 11 displays the climatological odds of storms tracking within 50 miles of each state along the Gulf and East Coasts along with the odds for the remainder of 2026. Landfall probabilities are well below their long-term averages. Probabilities for other Atlantic basin landmasses are available on our [website](#). Figures 39–41 display post 4-August 2026 probabilities of storms tracking within 50 miles of each coastal county from Texas to Maine for named storms, hurricanes and major hurricanes, respectively.

Given that landfall rates between 1880–2020 and 1991–2020 are similar for the continental US, we adjust all landfall rates relative to the 1991–2020 Atlantic west of 60°W ACE climatology. However, we use the longer 1880–2020 period for the climatological landfall statistics to increase the robustness of the historical landfall dataset. Also, storms near landfall are likely better observed than those farther east in the basin prior to the satellite era (e.g., mid-1960s). Slight differences in ACE west of 60°W between the two periods (73 for 1991–2020 vs. 66 for 1880–2020) are likely mostly due to improved observational technology in the more recent period.

Table 11: Probability of  $\geq 1$  named storm, hurricane and major hurricane tracking within 50 miles of each coastal state from Texas to Maine for the remainder of 2026. Probabilities are also provided for the 1880–2020 climatological average.

| State          | 2026 Probability     |              |                 | Climatological       |              |                 |
|----------------|----------------------|--------------|-----------------|----------------------|--------------|-----------------|
|                | Probability $\geq 1$ | event within | 50 miles        | Probability $\geq 1$ | event within | 50 miles        |
|                | Named Storm          | Hurricane    | Major Hurricane | Named Storm          | Hurricane    | Major Hurricane |
| Alabama        | 23%                  | 9%           | 3%              | 58%                  | 28%          | 8%              |
| Connecticut    | 7%                   | 2%           | <1%             | 22%                  | 8%           | 1%              |
| Delaware       | 7%                   | 2%           | <1%             | 23%                  | 6%           | 1%              |
| Florida        | 44%                  | 22%          | 10%             | 86%                  | 56%          | 29%             |
| Georgia        | 26%                  | 10%          | 2%              | 63%                  | 30%          | 6%              |
| Louisiana      | 28%                  | 14%          | 5%              | 66%                  | 38%          | 14%             |
| Maine          | 7%                   | 2%           | <1%             | 21%                  | 7%           | 1%              |
| Maryland       | 11%                  | 3%           | <1%             | 31%                  | 11%          | 1%              |
| Massachusetts  | 11%                  | 5%           | 1%              | 33%                  | 14%          | 3%              |
| Mississippi    | 20%                  | 10%          | 2%              | 53%                  | 28%          | 8%              |
| New Hampshire  | 6%                   | 2%           | <1%             | 18%                  | 6%           | 1%              |
| New Jersey     | 7%                   | 2%           | <1%             | 23%                  | 7%           | 1%              |
| New York       | 9%                   | 3%           | 1%              | 26%                  | 9%           | 2%              |
| North Carolina | 29%                  | 14%          | 2%              | 68%                  | 38%          | 8%              |
| Rhode Island   | 7%                   | 2%           | <1%             | 20%                  | 8%           | 1%              |
| South Carolina | 22%                  | 10%          | 3%              | 57%                  | 29%          | 8%              |
| Texas          | 25%                  | 13%          | 5%              | 61%                  | 36%          | 16%             |
| Virginia       | 17%                  | 6%           | <1%             | 46%                  | 20%          | 1%              |

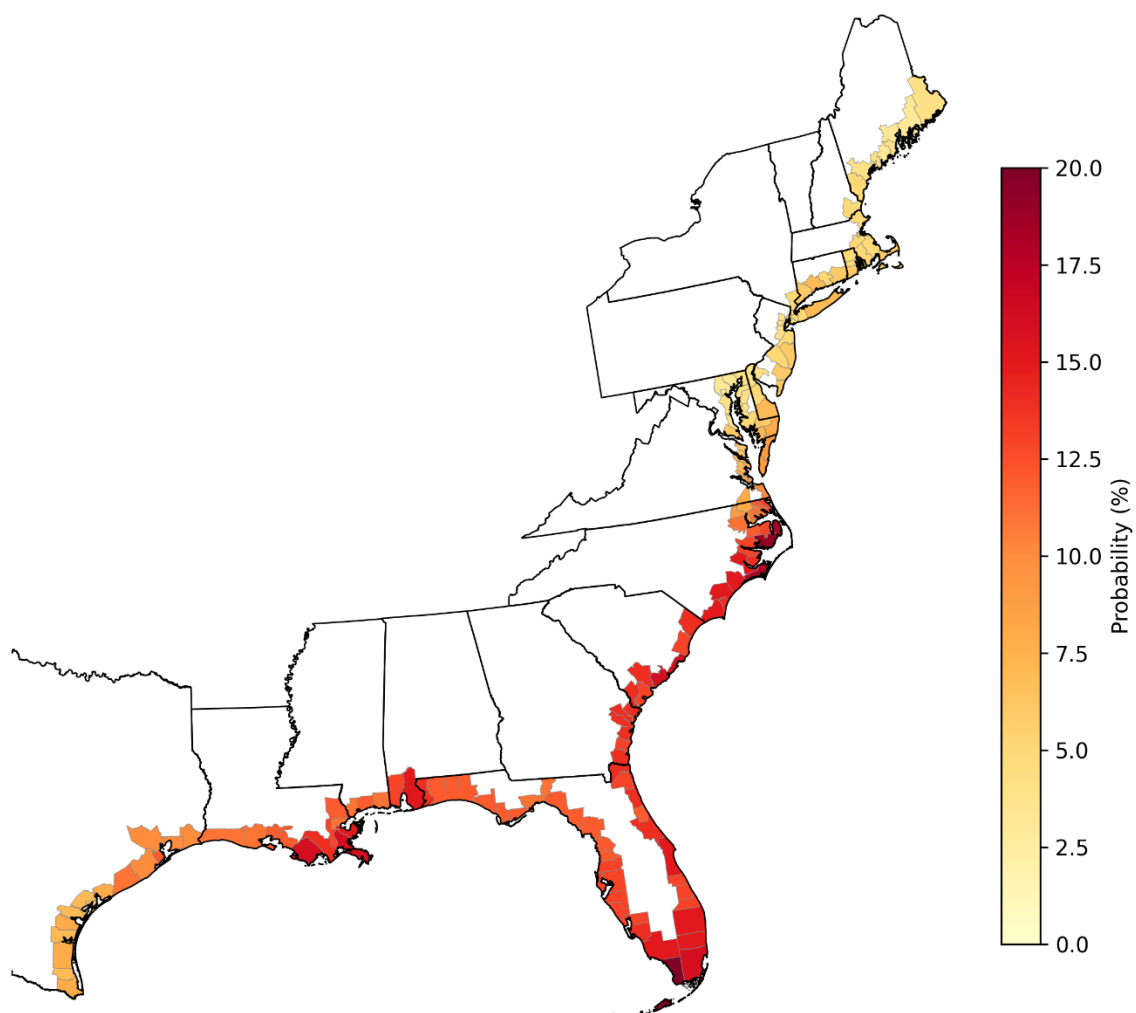


Figure 39: Odds of  $\geq 1$  named storm tracking within 50 miles of each coastal county from Texas to Maine for the remainder of 2026.

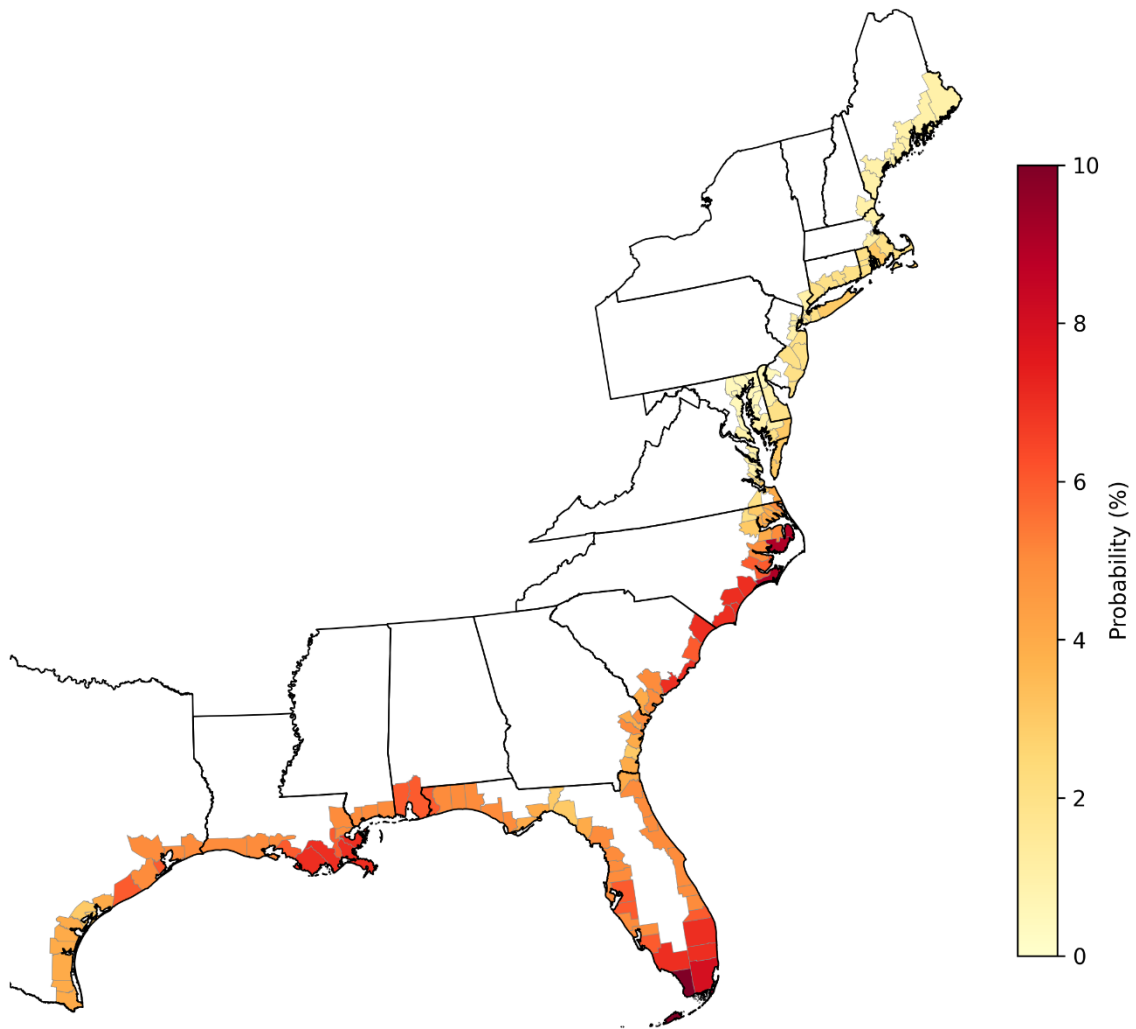


Figure 40: As in Figure 39 but for hurricanes.

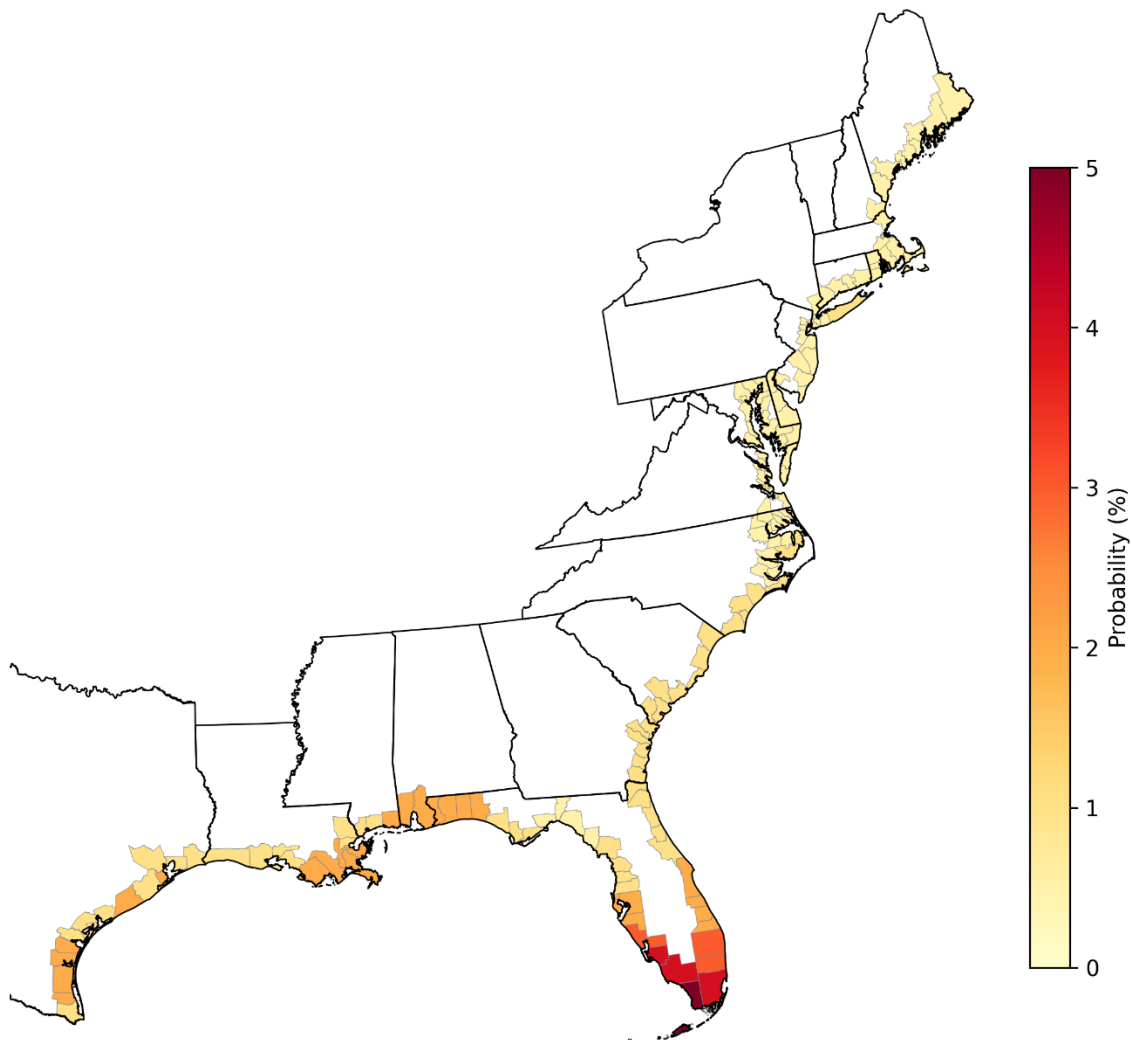


Figure 41: As in Figure 39 but for major hurricanes.

## 8 Summary

An analysis of a variety of different atmosphere and ocean measurements (through July) which are known to have long-period statistical relationships with the upcoming season's Atlantic tropical cyclone activity, as well as output from dynamical models, indicate that 2026 will have below-average activity due primarily to the anticipated powerful El Niño.

## 9 Forthcoming Updated Forecasts of 2026 Hurricane Activity

We will be issuing two-week forecasts for Atlantic TC activity during the climatological peak of the season from August–October, beginning today, Wednesday, 5 August and continuing every other Wednesday (19 August, 2 September, etc.). A verification and discussion of all 2026 forecasts will be issued on Tuesday, 24 November. All forecasts and verifications are available on our [website](#).

## 10 Verification of Previous Forecasts

CSU's seasonal hurricane forecasts have shown considerable improvement in recent years, likely due to a combination of improved physical understanding, adoption of statistical/dynamical models and more reliable reanalysis products. Figure 42 displays correlations between observed and predicted Atlantic hurricanes from 1984–2025, from 1984–2013 and from 2014–2025, respectively. Correlation skill has improved at all lead times in recent years, with the most noticeable improvements at longer lead times. While twelve years is a relatively short sample size, improvements in both modeling and physical understanding should continue to result in future improvements in seasonal Atlantic hurricane forecast skill. More detailed verification statistics are also available at: <https://tropical.colostate.edu/archive.html#verification>

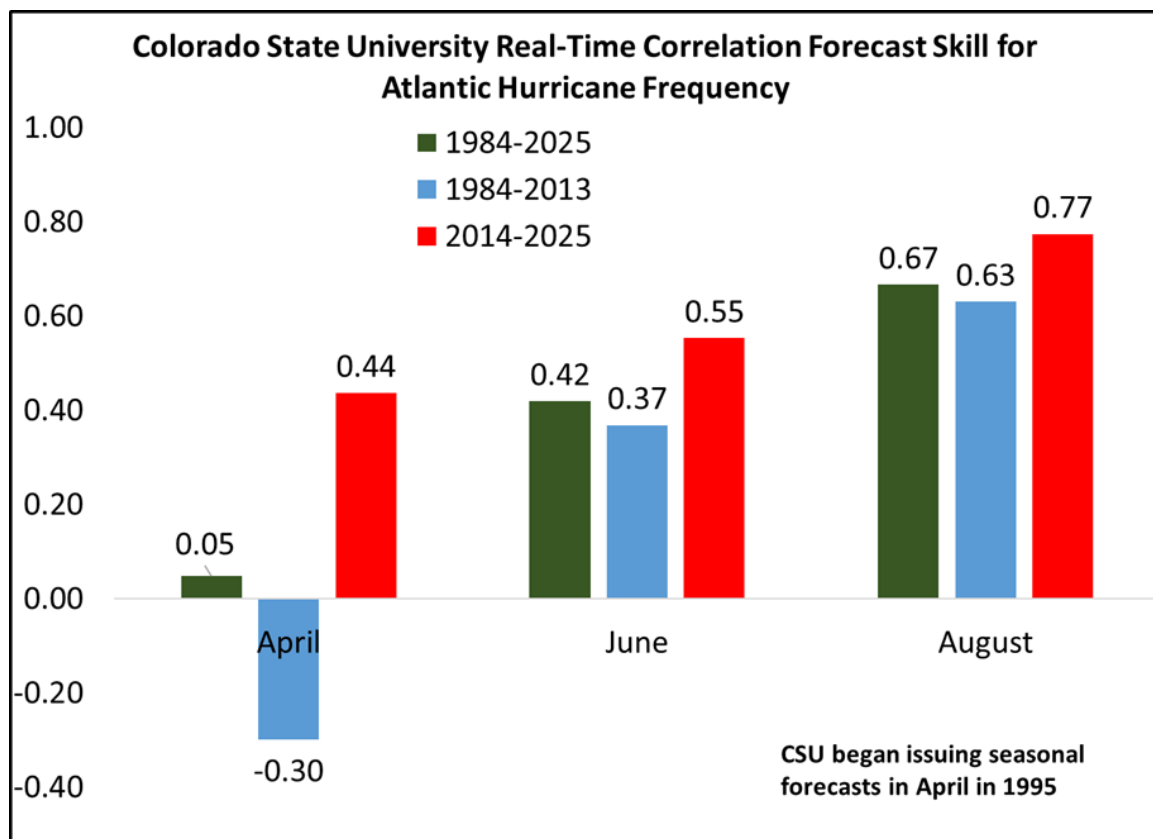


Figure 42: CSU's real-time forecast skill for Atlantic hurricanes using correlation as the skill metric. Correlation skills are displayed for three separate time periods: 1984–2013, 2014–2025 and 1984–2025, respectively.